Covert Movement, Chain Reduction, and Prosodic Grouping Effects

Abstract: This paper investigates a linear alternation governed by a prosodic constraint: Mandar (Austronesian, Sulawesi) shows a basic word order of VOS but routinely requires the absolutive argument to surface further to the left so that it can be contained in the same prosodic domain as the exponents of absolutive agreement. I show that this effect involves a type of phonologically conditioned chain reduction: the absolutive argument raises through a series of subject positions in the syntax, it is preferentially spelled out as low as possible, but it is also preferentially parsed into the same maximal phonological phrase as absolutive agreement—and when phonological constraints prevent this restriction from being satisfied by lower-copy spell-out, the absolutive argument is spelled out higher along its path. These results yield a set of interlocking arguments for the Copy Theory of Movement (Chomsky, 1993), the existence of covert A-movement (Bobaljik, 2002; Potsdam & Polinsky, 2012), the existence of prosodic grouping constraints (Deguchi & Kitagawa, 2002; Ishihara, 2003; Hirotani, 2005; Smith, 2005), and the possibility that chain reduction can be shaped by constraints on clause-level prosodic organization (Franks & King, 2000; Bošković, 2001)—suggesting that chain reduction must occur in the parallel and global phonological evaluation where prosodic structure is built (Selkirk, 2009, 2011).

Keywords:
Covert Movement, Chain Reduction, Syntax-Phonology, Distributed Morphology

1 Introduction

The Copy Theory of Movement holds that the surface forms of movement dependencies are established through a process of Chain Reduction along the PF branch (Chomsky, 1993). In the syntax, on this account, movement simply copies its targets into their derived positions to yield representations like (1a). In the analytical framework that has emerged around this view, these representations acquire their surface forms in the PF-level system of exponence (Halle & Marantz, 1993): at the derivational moment where the terminal nodes of the morphosyntax are given phonological form, lexical content is inserted into a single link to yield the output in (1b) (Saab, 2008, 2017; Muñoz-Pérez, 2018).

(1) *Chain Reduction*

a. SYNTAX:	$[_{\text{TP}}$	DP	AUX	$[_{\text{VP}}$	VERB	DP	]].
b. PF:	$[_{\text{TP}}$	John	was	$[_{\text{VP}}$	seen	Δ	]].

In the framework of Distributed Morphology (Halle & Marantz, 1993), lexical insertion is standardly positioned at the tail-end of the Morphological Module. This analysis makes it possible to understand how many aspects of Chain Reduction seem to be shaped by the systems of the morphology: for instance, individual links can be modified by morphological operations (Nunes, 2004; Harizanov, 2014a; Baker & Kramer, 2018) and multiple links can undergo insertion to satisfy morphological constraints (Landau, 2006; Jouitteau, 2012). But there are many patterns of Chain Reduction that seem to be shaped by phonological output constraints, which operate in a separate and later module in the standard architecture of DM (Franks, 1999; Bošković, 2001; Bjorkman, 2020). These results raise a network of questions about the structure of PF: what systems guide Chain Reduction, where does insertion occur, and how is information passed between the modules along that branch?

To push forward our understanding of these questions, this paper lays out a system where prosodic constraints derail a preference for insertion into lower positions in a chain. This is one that determines the linear position of the absolutive DP in Mandar, an Austronesian language of Central Indonesia. This language shows a basic word order of VSO and requires every finite clause to host absolutive agreement in T^0. This agreement typically lowers to surface after the highest auxiliary or verb (2a), and in root clauses where non-subjects move to the left periphery, it raises from T^0 to C^0 instead (2b). I will refer to the argument that triggers this agreement as the absolutive DP and underline it throughout.

(2) *Mandar*

a. [$_{\text{TP}}$ Nawaca **i** guru buku dionging].
read AGR teacher book yesterday
'The teacher read the book yesterday.'

b. [$_{\text{CP}}$ Pirang **i** nawaca buku dionging]?
when AGR read book yesterday
'When did she read the book yesterday?'

The absolutive DP preferentially falls into a three-word window with the host of absolutive agreement. In three-word strings of verbs and arguments, the absolutive DP can always satisfy this constraint from its base position, and as a result, it always surfaces there. But when additional words are introduced across the clause, it is often forced to surface farther to the left to fall into this window—appearing at the edge of the *voice*P (3a) or higher in the extended TP (3b). I will refer to this as Absolutive Preposing below.

(3) *Absolutive Preposing*

a. [$_{\text{TP}}$ Nawaca-waca **i** **buku** guru △ dionging].
RED-read AGR book teacher yesterday
'The teacher sorta read the book yesterday.'

b. [$_{\text{CP}}$ Jas sapa **i** **buku** nawaca △ manini]?
time what AGR book read later
'What time will they read the book later?'

The first task of the paper is to show that this interaction implicates Chain Reduction: the absolutive DP raises to the edge of the *voice*P and then into the TP in the syntax (6a), it is preferentially exponed in the lowest position along this path (6b), and it is realized higher whenever that allows it to satisfy this constraint (6c). To this end, I first show that the absolutive DP typically undergoes covert A-movement (Potsdam & Polinsky, 2012; Kishimoto, 2013; Deal, 2017): no matter its linear position, it c-commands the ergative DP for the purposes of Variable Binding and Condition C. I then show that independent pressures can force realization of the absolutive DP at the edge of the *voice*P and in a higher position in the TP. These facts reveal that the absolutive DPs which descriptively undergo preposing are simply realized in syntactic positions they move through covertly—rather than undergoing any type of phonological movement that draws them to prosodically-defined positions in service of phonological output constraints (Halpern 1995; Bennett *et al.* 2016; Kusmer 2019; see also Richards 2016; Branan 2018; Clemens 2019).

(4) *Variation in Spell-Out*

a. SYNTAX: $[_{\text{TP}}$ DP$_{\text{ABS}}$ V $[_{\textit{voice}\text{P}}$ DP$_{\text{ABS}}$ DP$_{\text{ERG}}$ $[_{\textit{voice}\text{P}}$ DP$_{\text{ABS}}$]]].

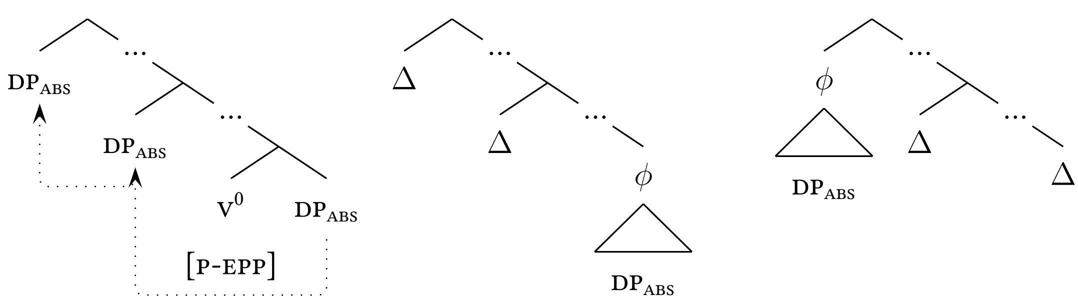
b. USUAL PF: $[_{\text{TP}}$ Δ V $[_{\textit{voice}\text{P}}$ Δ DP$_{\text{ERG}}$ $[_{\textit{voice}\text{P}}$ DP$_{\text{ABS}}$]]].

c. SPECIAL PF: $[_{\text{TP}}$ DP$_{\text{ABS}}$ V $[_{\textit{voice}\text{P}}$ Δ DP$_{\text{ERG}}$ $[_{\textit{voice}\text{P}}$ Δ]]].

The second task of the paper is to integrate this effect with the architecture of the PF branch. To capture the preference for low insertion, I propose that heads can bear features that mark individual positions as preferential targets for insertion—surfacing low in the Mandar clause to drive low realization in the default case (5a)-(5b). To capture the facts of Absolutive Preposing, I propose that insertion must proceed in the constraint-based and optimizing phonological calculus where prosodic structure is built (Selkirk 2009, 2011). In this module, it is guided by three types of constraints: the pressures that guide Chain Reduction, the pressures that force the faithful execution of instructions from the morphosyntax, and output-oriented phonological constraints. This last family of constraints can outrank and derail the pressure to follow the instructions of the morphosyntax to force Absolutive Preposing: the exceptional realization of higher copies in (5c).

(5) *Chain Reduction*

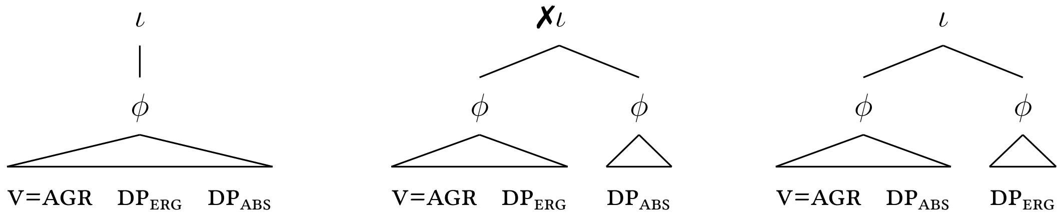
a. *Morphosyntax* b. *Low Insertion* c. *High Insertion*



The third task of the paper is to show that the pressure behind Absolutive Preposing is a violable constraint on Prosodic Grouping (Deguchi & Kitagawa 2002; Ishihara 2003; Hirotani 2005; Smith 2005; see also Richards 2016; Branan 2018): one that requires the absolutive DP to fall into a phonological phrase with the exponent of absolutive agreement. When the base position of the absolutive DP falls into a three-word window with the host of absolutive agreement, it is possible to satisfy this constraint by inserting the absolutive DP in the lowest position along its chain (6a). When that position should fall outside of the three-word window with the host of agreement, this constraint disfavors output representations like (6b)—and ranked above the pressure to follow the instructions of the morphosyntax, it forces insertion into the next-lowest position along the chain that allows the absolutive DP to satisfy this constraint (6c).

(6) *Output Optimization*

a. *Default Spell-Out* b. *Prosodic Grouping* c. *Higher Spell-Out*



The case proceeds in the following way. Section 2 lays out the clausal syntax of Mandar and introduces Absolutive Preposing. Section 3 shows that the absolutive DP raises through two positions in the syntax, identified as SPEC,*voice*P and SPEC,ASPP, and shows that each step can be independently rendered overt. Section 4 lays out a theory of Chain Reduction to capture these effects. Section 5 returns to the phonology of Absolutive Preposing and shows that it optimizes for the Grouping Constraint. Section 6 concludes.

2 Background

Mandar is spoken by roughly 400,000 people in the Indonesian province of West Sulawesi. It belongs to the South Sulawesi Subfamily, a primary branch of Malayo-Polynesian (Esser, 1938; Mills, 1975), and it is the subject of several descriptive works by the Language Office of South Sulawesi (Pelenkahu *et al.*, 1983; Sikki *et al.*, 1987; Muthalib & Sangi, 1991). This paper describes standard Mandar, which is spoken along the south coast of West Sulawesi (Grimes & Grimes, 1987). It presents data gathered primarily in elicitation with Jupri Talib, a speaker whom I have worked with continually since 2018. This data been cross-checked over regular trips to the field by interaction with the wider community and explicit verification with two native speaker linguists, Sitti Sapiah and Hairuddin. I believe that it is representative of standard Mandar and replicable in many dialects and languages nearby.[1]

[1] GLOSSES: ABS: absolutive, ANTIP: APPL: applicative, ERG: ergative, GEN: genitive; SG/PL: singular/plural.

2.1 Clause Structure

Mandar shows a consistent and rigid basic word order of V-EXT-INT-GOAL (verb > external argument > internal argument > applied argument). The verb undergoes head-movement to *voice*^{0} and locative and temporal adjuncts surface after the *voice*P. Arguments typically surface between the verb and *voice*P-final adjuncts within the *v*P, which forms a constituent for various syntactic processes like ellipsis and predicate clefting and also typically forms a consistent in the surface phonology as well (Brodkin, 2025a). In the *v*P, they take a fixed order in V-N-N strings of three prosodic words: the EXT precedes the INT (7a) and the EXT and INT precede the GOAL (7b).

(7) *The Mandar Clause*

a. Naita i [$_{vP}$ **dottor** **guru**] dionging.
   see doctor teacher yesterday.
   'The doctor saw the teacher yesterday.'

b. Uyoloang i [$_{vP}$ **dottor** **guru**] dionging.
   I showed doctor teacher yesterday
   'I showed the doctor to the teacher yesterday.'

2.1.1 The Shape of the Verb

Transitive and ditransitive verbs always show ergative agreement with the EXT (8a). Ditransitive verbs invariably carry applicative suffixes as well (8b). Absolutive agreement targets the INT when the verb is transitive and the GOAL when the verb is ditransitive.

(8) *The Transitive and Ditransitive*

a. **U**-timbangngi i yau do'ayu digena'.
   1ERG-weigh 3ABS 1SG vegetable earlier
   'I weighed the vegetables earlier.'

b. **Na**-timbangngi-**ang** a' dottor paulli digena'.
   3ERG-weigh-APPL 1ABS doctor medicine earlier
   'The doctor weighed the medicine for me earlier.'

Ergative agreement is absent from unergative verbs, which host the infix *-um-*, and from unaccusative verbs, which surface bare (9a). It is also absent from antipassive verbs, which host the prefix *mang-*. Antipassive verbs always take indefinite internal arguments

(9b), and when they take applied arguments, they require them to be indefinite as well (9c). Absolutive agreement targets the EXT when the verb is antipassive.

(9) *Unergatives, Unaccusatives, and Antipassives*

a.

L<***um***>amba	a’	yau	di bongi	na	**tanda**	a’	malimang.
UNERGATIVE-go	1ABS	1SG	at night	and	arrive	1ABS	morning

‘I went last night and arrived this morning.’

b.

Mat-timbang	a’	yau	do’ayu	digena’.
ANTIP-weigh	1ABS	1SG	vegetable	earlier

‘I weighed some vegetables earlier.’

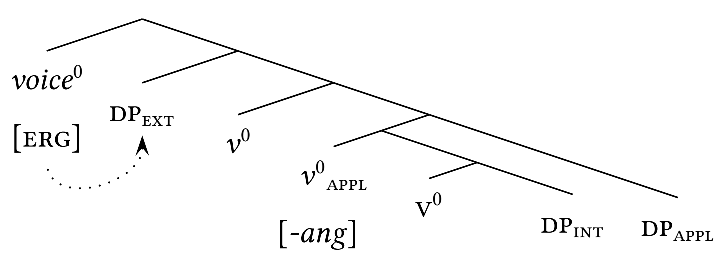
c.

Mat-timbang-**ang**	a’	paulli	sanaeke	digena’.
ANTIP-weigh-APPL	1ABS	medicine	children	earlier

‘I weighed some medicine for children earlier.’

I propose that the *voice*P shows the syntax in (10). The EXT is base-merged in SPEC,*v*P, the GOAL in a rightward SPEC,*appl*P, and the INT as the sister of V. The applicative suffix sits in *appl*^{0} and ergative agreement and the antipassive prefix sit in *voice*^{0}. I assume that ergative agreement proceeds through the syntactic process of AGREE (Chomsky, 2001), which endows its target with abstract Case (Vergnaud, 1977) and licenses the external arguments of transitive and ditransitive verbs in the *v*P (Aldridge, 2004; Wiltschko, 2006).

(10) *The VoiceP*



2.1.2 Auxiliaries and Verb-Movement

Mandar has a class of preverbal auxiliaries, listed in (11), that mark distinctions of polarity, tense, and aspect. These elements are rigidly ordered and typically fail to form words.

(11) *The Auxiliary System*

Σ^0		TENSE^0		MOD^0		ASP$^0_{\text{HAB/EXP}}$		ASP$^0_{\text{COMP}}$	
ndang	not	*pura*	PAST	*mala*	can/may	*byasa*	usually	*yari*	COMPLETE
da	don’t	*na*	FUT	*pali*	have time	*rua*	ever	*lesse’*	INCOMPLETE

In the clauses that contain auxiliaries, the verb remains in the *voice*P (12a). In clauses that lack auxiliaries, the verb raises into the middle field instead (12b). I propose that it is drawn to this position by head-movement to Σ^0, the usual position of sentential negation. In clauses that contain auxiliaries, I also assume that the highest auxiliary raises to Σ^0.

(12) *Verb-Movement*

a.

$[_{\Sigma\text{P}}$	Ndang	i	yari	$[_{voice\text{P}}$	u-ita	dottor	dionging]].
	not	3ABS	COMPLETE		1ERG-see	doctor	yesterday

'I didn't see the doctor yesterday.'

b.

$[_{\Sigma\text{P}}$	U-ita	i	$[_{voice\text{P}}$	__	dottor	dionging]].
	1ERG-see	3ABS			doctor	yesterday

'I saw the doctor yesterday.'

2.1.3 Absolutive Agreement and Viewpoint Aspect

Absolutive agreement sits above the auxiliaries in T^0. It forms portmanteaux with viewpoint aspect clitics, which sit below T^0 in $\text{ASP}^0_{\text{VIEW}}$ and typically raise to T^0. These elements follow the verb in verb-initial clauses (13a) and the highest auxiliary in AUX-initial clauses (13b). I propose that they reach this position via postsyntactic lowering to Σ^0.

(13) *Absolutive Agreement and Viewpoint Aspect*

a.

$[_{\Sigma\text{P}}$	U-ita	mi	$[_{voice\text{P}}$	setang	dio]].
	1ERG-see	3ABS.NOW		demon	there

'Now I see the demon there.'

b.

$[_{\Sigma\text{P}}$	Rua	mi	$[_{voice\text{P}}$	u-ita	setang	dio]].
	once	3ABS.NOW		1ERG-see	demon	there

'Now I've seen the demon there.'

These clitics follow adjunct WH-phrases when they raise to SPEC,CP (14). In these cases, I propose that T^0 undergoes syntactic head-movement to C^0, bleeding lowering to Σ^0.

(14) *Head-Movement to C*

a.

$[_{\text{CP}}$	Myapa	mi	$[_{\Sigma\text{P}}$	mu-sa'ding]]?
	how	3ABS.NOW		2ERG-feel

'How do you feel now?'

b.

$[_{\text{CP}}$	Pirang	pai	$[_{\Sigma\text{P}}$	na	mu-ita]]?
	when	3ABS.YET		FUTURE	2ERG-see

'When will you see it?'

2.2 The Three-Word Window

The absolutive DP takes its usual position in the EXT-INT-GOAL string when that position falls into a three-word window with the host of absolutive agreement. As a result, it always retains its base position in three-word strings: when it is the INT, it follows the EXT, and when it is the GOAL, it follows the INT (15). To illustrate these effects, I will mark words with accents and mark this three-word window with braces in the examples below.

(15) *Absolutive Arguments: Regular Order*

a.

$\{_\phi$	Na-báca	i	gúru	**búku**	}	dionging.
	3ERG-read	3ABS	teacher	book		yesterday

'The teacher read the book yesterday.'

b.

$\{_\phi$	Na-báca-ngang	i	búku	**gúru**	}	dionging.
	3ERG-read-APPL	3ABS	book	teacher		yesterday

'They read the book to the teacher yesterday.'

The absolutive DP also retains its usual position when the overall distribution of words prevents it from falling into a three-word window with the host of agreement. It thus remains in place when it contains three words (16a) or when the host of agreement contains three words (16b). The absolutive DP also remains in place whenever it and the host of agreement collectively contain more than three words, as in (16c).

(16) *Prosodic Disruptions*

a.

Na-báca	i	gúru	**búku**	**ílmu**	**bása**	digéna'.
3ERG-read	3ABS	teacher	book	science	language	earlier

'The teacher read the language science book earlier.'

b.

Mé-na-báca-báca	i	gúru	búku	digéna'.
JUST-3ERG-RED-read	3ABS	teacher	book	earlier

'The teacher just sorta read the book earlier.'

c.

Na-báca-báca	i	gúru	**búku**	**linguístik**	digéna'.
3ERG-RED-read	3ABS	teacher	book	linguistic	earlier

'The teacher sorta read the linguistics book earlier.'

2.2.1 Movement to SPEC,*Voice*P

When the absolutive DP and the host of agreement collectively contain two or three words, we see a different effect when the base position of the absolutive DP should fall outside the three-word window with the host of absolutive agreement. In strings of the order V-EXT-INT, we can force the base position of the INT to fall outside of this window by adding words in three places: in the INT (17a), in the verb (17b), and in the EXT (17c).

(17) *Expulsion from the Window*

a.							
{$_\phi$	Na-báca-ngang	a’	gúru	}	**búku**	**linguístik**	diónging.
	3ERG-read-APPL	1ABS	teacher		book	linguistic	yesterday

‘The teacher read me the linguistics book yesterday.’

b.						
{$_\phi$	**Na-báca-báca-ngang**	a’	gúru	}	búku	diónging.
	3ERG-RED-read-APPL	1ABS	teacher		book	yesterday

‘The teacher sorta read me the book yesterday.’

c.							
{$_\phi$	Na-báca-ngang	a’	**gúru**	**linguístik**	}	búku	diónging.
	3ERG-read-APPL	1ABS	teacher	linguistic		book	yesterday

‘The linguistics teacher read me the book yesterday.’

When the INT is absolutive in this context, it takes a position that follows the verb (18). I will refer to this as Absolutive Preposing and mark the position that it escapes with the symbol ∆ below. I will refer to the pressure that drives it as the Grouping Constraint.

(18) *Absolutive Preposing*

a.								
{$_\phi$	Na-báca	i	**búku**	**linguístik**	}	gúru	∆	diónging.
	3ERG-read	3ABS	book	linguistics		teacher		yesterday

‘The teacher read the linguistics book yesterday.’

b.							
{$_\phi$	**Na-báca-báca**	i	búku	}	gúru	∆	diónging.
	3ERG-RED-read	3ABS	book		teacher		yesterday

‘The teacher sorta read the book yesterday.’

c.								
{$_\phi$	Na-báca	i	búku	}	**gúru**	**linguístik**	∆	diónging.
	3ERG-read	3ABS	book		teacher	linguistic		yesterday

‘The linguistics teacher read the book yesterday.’

In this environment, the absolutive DP is realized in a consistent syntactic position: it does not simply shift the minimum possible distance to the left. We can see this effect from the behavior of the GOAL, which follows the INT and the EXT whenever it is not the absolutive DP (19a). When the GOAL is absolutive and when its base position follows two overt DPs, it never satisfies the Grouping Constraint by shifting across one DP. In verb-initial clauses where the absolutive GOAL and the verb collectively contain two or three words, rather, it shifts to a position that immediately follows the verb (19b).

(19) *The Landing Site*

a. M-álli-ang i <u>gúru</u> búku **sanaéke** digéna'.
ANTIP-buy-APPL 3ABS teacher book kid earlier
'The teacher was buying books for kids earlier.'

b. {$_\phi$ Na-álli-ang i <u>**sanaéke**</u> gúru } búku △ digéna'.
3ERG-buy-APPL 3ABS kid teacher book earlier
'The teacher bought books for the kid earlier.'

This target position falls at the edge of the *voice*P. We can see this fact by introducing auxiliaries, which prevent the verb from raising to Σ^0. In that environment, the absolutive agreement and viewpoint aspect clitics lower to the highest auxiliary and force it to form a word. When the absolutive DP falls into a three-word window with that word from its base position, it remains in place (20a). When it can only fall into that window by shifting to the left, it takes a position immediately above the verb (20b). It takes this position even if it could also satisfy the Grouping Constraint by immediately following the verb (20c).

(20) *Beneath Auxiliaries*

a. {$_\phi$ Ndáng i rua na-báca <u>**búku**</u> } diónging.
not 3ABS ever 3ERG-read book yesterday
'She never read the book yesterday.'

b. {$_\phi$ Ndáng i rua <u>**búku**</u> <u>**linguístik**</u> } na-báca △ diónging.
not 3ABS ever book linguistics 3ERG-read yesterday
'She never read the linguistics book yesterday.'

c. {$_\phi$ Ndáng i rua <u>**búku**</u> **na-báca** } gúru △ diónging.
not 3ABS ever book 3ERG-read teacher yesterday
'The teacher never read the book yesterday.'

2.2.2 Movement to SPEC,AspPP

When absolutive agreement raises to C^0, we see an analogous effect. When the absolutive DP can satisfy the Grouping Constraint from its base position, it remains in place (21a). When it can only satisfy that constraint from SPEC,*voice*P, it surfaces there (21b).

(21) *Grouping from the CP*

a. {$_\phi$ Pírang i na-báca <u>**búku**</u> } diónging?
when 3ABS 3ERG-read book yesterday
'When did she read the book yesterday?'

b. {$_\phi$ Pírang i na-báca **búku** } gúru △ diónging?
when 3ABS 3ERG-read book teacher yesterday
'When did the teacher read the book yesterday?'

When the absolutive DP cannot satisfy the Grouping Constraint from SPEC,*voice*P, we see a second step: when the absolutive DP and the WH-phrase that hosts agreement collectively form three words, the absolutive DP surfaces above the verb in Σ^0 (22a)-(22b). We see the same effect when the absolutive DP and the hosting WH-phrase collectively form two words and SPEC,*voice*P is separated from the WH-phrase by a two-word verb (22c).

(22) *Movement to [Spec,Asp$_{View}$]*

a. {$_\phi$ **Pírang** i **búku linguístik** } na-báca △ diónging?
when 3ABS book linguistics 3ERG-read yesterday
'When did she read the linguistics book yesterday?'

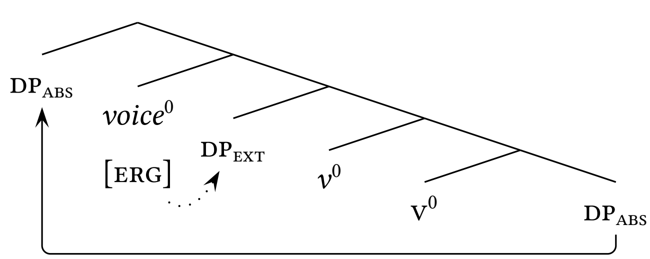
b. {$_\phi$ **Jáng sangápa** i **búku** } na-báca △ diónging?
time what 3ABS book 3ERG-read yesterday
'What time did she read the book yesterday?'

c. {$_\phi$ Pírang i búku } **na-báca-báca** △ dionging?
when 3ABS book 3ERG-RED-read yesterday
'When did she sorta read the book yesterday?'

2.3 High Absolutive Syntax

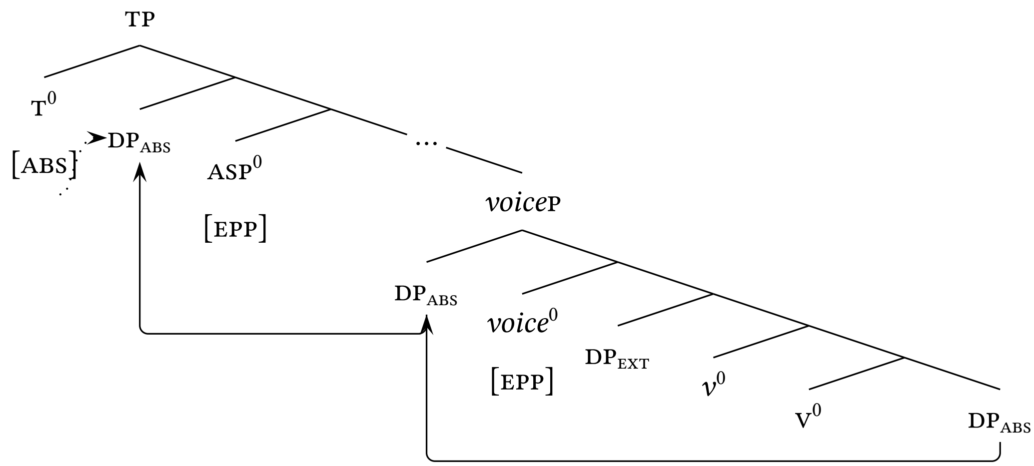
To understand how this type of movement might occur, we can turn to the system of agreement. When *voice*^{0} triggers ergative agreement with the EXT, I propose that it renders it inactive and invisible to all following operations in the A-syntax (Chomsky, 2001). If *voice*^{0} probes again to attract the highest argument in *v*P at this stage, as a result, it should always encounter the argument that will become the absolutive DP: the transitive INT or the ditransitive GOAL. To capture the first step of Absolutive Preposing, I thus propose the step of syntactic movement in (23): *voice*^{0} attracts the absolutive DP to SPEC,*voice*P.

(23) *The Link to Licensing*



To capture the second step of Absolutive Preposing, I propose that the absolutive DP is then attracted to a higher A-position in the extended TP: the specifier of $\text{ASP}_{\text{VIEW}}{}^0$. I propose that it moves to this position to satisfy an A-probe that draws it up from SPEC,*voice*P. The result is a second step of A-movement that positions it to receive abstract Case from T^0.

(24) *The Higher Position*



In the clauses that show Absolutive Preposing, this analysis allows us to understand the absolutive DP in much the same way as a nominative subject: it raises to the highest A-position in the clause as part of a system that licenses it with the highest clause-internal source of Structural Case. In regional perspective, this analysis establishes a structural parallel between the absolutive DP in Mandar and the arguments identified as *pivots* or *subjects* in many other languages of the Western Austronesian area, which are taken to raise into the highest A-position in the clause by the mainstream literature on the region (Guilfoyle *et al.*, 1992; Aldridge, 2004, 2008, 2012; Nomoto, 2013; Erlewine, 2018; Nie, 2020; Hsieh, 2020; Ting, 2023; Erlewine & Sommerlot, 2025). In wider theoretical perspective, it also establishes Mandar within the class of ergative languages that shows High Absolutive Syntax, in the sense of Coon *et al.* 2014 (see also Aldridge 2004; Legate 2006; Ershova 2023).

3 The Case for Chain Reduction

To integrate the facts of Absolutive Preposing with the syntax of agreement, I thus propose that the absolutive DP consistently raises to SPEC,*voice*P and then further to SPEC,*voice*P. In the default case, I propose, these steps are covert—but when the Grouping Constraint interferes, we see overt movement of the absolutive DP. On this analysis, the absolutive DP will always raises past the highest position of the verb in the narrow syntax—placing

Mandar in a class with many languages whose subjects typically surface preverbally and setting it apart from the verb-initial languages that require the verb to raise past the highest subject position in the syntax (for instance, many languages of the Philippines (Rackowski 2002; Aldridge 2004; Nie 2020; Hsieh 2020; but *cf.* Guilfoyle *et al.* 1992) and the High-Absolutive languages of the Mayan family (Coon *et al.*, 2014; Royer, 2025)).

3.1 Covert Movement

The case for this analysis begins in a range of reduced constructions that lack both absolutive agreement and T^0. One of these is selected by the subordinator *mau* "though," and I will refer to it as a though-complement below. This construction contains all the structure beneath the head ASP$^0_{\text{VIEW}}$, and as a result, I take it to project up to ΣP. It is able to host auxiliaries (25a) and when it does not, it shows verb-raising to Σ^0 (25b)

(25) *Though-Complements*

a. **Mau** [$_{\Sigma P}$ ndang rua [$_{voiceP}$ na-ita dio di perpus]],
though not ever 3ERG-see there in library
'Though never seen in the library,'

b. **Mau** [$_{\Sigma P}$ na-ita [$_{voiceP}$ ___ dio di perpus]],
though 3ERG-see there in library
'Though seen in the library,'

Though-complements can contain overt absolutive DPs. Unlike the clauses that show absolutive agreement, however, they prohibit Absolutive Preposing. As a result, they always require absolutive DPs to take their usual positions in the V-EXT-INT-GOAL string: absolutive INTS must follow the EXT (26a) and absolutive GOALS must follow the EXT and INT (26b). In this context, I take absolutive DPs to receive Case from the subordinating P^0.

(26) *Absolutive Arguments without Agreement*

a. {$_{\phi}$ **Mau** na-íta gúru linguístik } **dóttor** dío,
though 3ERG-see teacher linguistics doctor there
'Though the linguistics teacher saw the doctor there,'

b. {$_{\phi}$ **Mau** na-bé-ngang gúru búku } **dóttor** dío,
though 3ERG-give-APPL teacher book doctor there
'Though the teacher gave the book to the doctor there,'

The case for covert movement follows from the patterns of variable binding that emerge in this context. Mandar has three quantifiers that express universal quantification: *nasang* "every," *ianasanna* "each," and *inggannanna* "every." In non-finite clauses, *nasang* follows its associated DP (27a), while *ianasanna* and *inggannanna* surface prenominally (27b).

(27) *Universal Quantification: Nasang*

a.

Mau	u-baca	**buku**	**nasang**	dionging,
hough	1ERG-read	book	every	yesterday

'Though I read every book yesterday,'

b.

Mau	u-baca	**ianasanna**	**buku**	sola	**inggannanna**	sanaeke,
though	1ERG-read	each	book	with all		kid

'Though I read each book with all the kids,'

When the ergative EXT is quantified in this way, it can bind variables in the INT and GOAL (28a). When the non-absolutive INT is quantified, however, it can never bind into the EXT or the GOAL (28b). These facts suggest that variable binding between coarguments requires c-command in Mandar (and see also Chomsky 1981; Reinhart 1983): it can proceed from the EXT into the INT and GOAL, tracing c-command in the *voice*P, but it cannot proceed into the EXT or the GOAL from the non-absolutive INT, which sits beneath them.

(28) *Variable Binding: C-Command*

a.

Mau	na-kiring-ang	**panulis**	**nasang**	buku-nna	kindo'-na	dionging,
though	3ERG-send-APPL	author	every	book-her	mom-her	yesterday

'Though every author$_i$ sent her$_{i,j}$ book to her$_{i,j,k}$ mom yesterday,'

b.

Mau	na-kiring-ang	panulis-na	buku	**nasang**	penerbit-na	dionging,
though	3ERG-send-APPL	author-its	book	every	publisher-its	yesterday

'Though its$_i$ author sent its$_{i,j}$ publisher every book$_{*i,*j,k}$ book yesterday,'

When the absolutive DP is quantified, however, it can bind into every other argument in the *voice*P. Even though it takes its usual position in the V-EXT-INT-STRING, in other words, the facts of variable binding suggest that it undergoes covert A-movement to a position above the *voice*P (and for similar effects elsewhere, see Potsdam & Polinsky 2012; Polinsky & Potsdam 2013; Kishimoto 2013; Deal 2017). This step allows the absolutive INT to bind into the EXT (29a) and the absolutive GOAL to bind into the EXT and INT (29b).

(29) *Variable Binding: Covert Movement*

a.

Mau	na-kiring	panulis-na	**buku**	**nasang**	dionging,
though	3ERG-send	author-its	book	every	yesterday

'Though its$_i$ author sent every book$_{i,j}$ yesterday,'

b.

Mau	na-kiring-ang	penerbit-na	buku-nna	**panulis**	**nasang**	dionging,
though	3ERG-send-APPL	publisher-her	book-her	author	every	yesterday

'Though her$_i$ publisher sent her$_{i,j}$ book to every author$_{i,\ j,\ k}$ book yesterday,'

We can make the same argument with Condition C. When the EXT is a pronoun, it imposes Condition C effects on the non-absolutive INT: it cannot co-refer with R-expressions in that DP (30a). When the INT is not absolutive, it does not impose Condition C effects on the EXT: when it is a pronoun, it can co-refer with R-expressions in the EXT (30b). I take these facts to suggest that pronominal coreference is sensitive to C-command in Mandar in just the same way: pronouns cannot co-refer with DPs in their C-command domains.

(30) *Condition C: C-Command*

a.

Mau	na-jolo-ang	***pro***	buku-nna	guru	mai	dionging,
though	3ERG-show-APPL	he	book-of	teacher	to me	yesterday

'Though he$_i$ showed me the teacher$_{*i,j}$'s books yesterday,'

b.

Mau	na-jolo-ang	sola-na	guru	***pro***	mai	dionging,
though	3ERG-send-APPL	friend-of	teacher	him	to me	yesterday

'Though the teacher$_i$'s friends showed him$_{i,j}$ to me yesterday,'

The absolutive DP escapes Condition C effects from all other arguments in the *voice*P: when the INT is absolutive, it can contain R-expressions coindexed with a pronominal EXT (31a). The absolutive DP also imposes Condition C effects on everything else in the *voice*P: when the INT is a pronoun, it cannot corefer with R-expressions in the EXT (31b). The result is a second argument for covert A-movement of the absolutive DP: to capture the facts of Condition C, the absolutive DP must raise above all other arguments in the *voice*P.

(31) *Condition C: Covert Movement*

a.

Mau	na	na-baca	***pro***	[$_{\text{INT}}$ buku	na-alli	guru	digena']	manini,,
though	FUTURE	3ERG-read	he	book	3ERG-buy	teacher	earlier	later

'Though he$_i$ will read the book the teacher$_{i,j}$ bought earlier later,'

b.

Mau	na-ita	[$_{\text{EXT}}$ sola-na	guru	na	dottor]	***pro***	dionging,
though	3ERG-see	friend-of	teacher	and	doctor]	him	yesterday

'Though the teacher$_i$ and the doctor's friends saw him$_{*i,j}$ yesterday,'

3.2 Overt Movement

These facts are matched by a second line of evidence for syntactic movement to SPEC,*voice*P and then SPEC,ASP$_{\text{VIEW}}$P. When the absolutive DP hosts *ianasanna* “each” or *inggannanna* “all,” it always surfaces in SPEC,*voice*P in though-complements. This step places absolutive INTS above the EXT (32a) and absolutive GOALS above the EXT and INT (32b).

(32) *Overt Movement*

a.

Mau	na-nilai	[*voice*P	**inggannanna**	**ujiang**	guru	Δ	]	sangallo,
though	3ERG-grade		all	test	teacher			in a day

‘Though the teacher graded all the tests in a day,’

b.

Mau	na-baca-ngang	[*voice*P	**ian.**	**sanaeke**	guru	buku	Δ	]	sangallo,
though	3ERG-read-APPL		each	kid	teacher	book			in a day

‘Though the teacher read the book to each kid in a day,’

This linear shift is obligatory and independent from the Grouping Constraint. In the presence of auxiliaries, it places its targets above the verb in *voice*^{0} (33). Like Preposing, it never places the absolutive DP after the verb when the verb does not raise to Σ^0.

(33) *Movement to [Spec,[Voice]*

a.

Mau	rua	[*voice*P	**ing.**	**ujiang**	na-nilai	guru	Δ	]	sangallo,
though	once		all	test	3ERG-grade	teacher			in a day

‘Though the teacher once graded all the tests in a day,’

b.

Mau	rua	[*voice*P	**ing.**	**ujiang**	**ilmu**	**basa**	na-nilai	Δ	]	sangallo,
though	once		all	test	science	language	3ERG-grade			in a day

‘Though she once graded all the linguistics tests in a day,’

This shift exclusively targets the absolutive DP. When the same quantifiers surface in other arguments, like the non-absolutive INT, they retain their positions in the *v*P (34).

(34) *Only the Absolutive*

a.

Mau	yari	[*voice*P	**ing.**	**sanaeke**	u-jolo-ang	**ian.**	**buku**	Δ	]	digena’,
though	COMPLETE		all	kid	1ERG-show-APPL	each	book			earlier

‘Though I showed all the kids each book earlier,’

b.

Mau	byasa	[*voice*P	**ian.**	**sanaeke**	u-baca-ngang	**ing.**	**buku**	Δ	]	dini,
though	usually		each	kid	1ERG-read-APPL	all	book			here

‘Though I usually read each kid all the books here,’ When we introduce viewpoint aspect in this construction, we see a second step. It is possible to add the viewpoint aspect clitics to though-complements of this type, and in this context, they lower to Σ^0 and follow the highest auxiliary or the raised verb (35). In their presence, I propose that though-complements exceptionally project to ASP$_{\text{VIEW}}$P.

(35) *Adding Structure*

a.

Mau	[$_{\text{ASPP}}$	yari	**mo**	u-nilai	<u>ujiang</u>],
though		COMPLETE	NOW	1ERG-grade	test

‘Though now I’ve graded the tests,’

b.

Mau	[$_{\text{ASPP}}$	na-baca-ngang	**mo**	___	guru	buku	<u>sanaeke</u>],
though		3ERG-read-APPL	NOW		teacher	book	kid

‘Though now the teacher has read the books to the kids,’

When we introduce prenominal quantifiers to the absolutive DP in this context, it begins to surface in SPEC,ASP$_{\text{VIEW}}$P. In verb-initial clauses, it precedes the verb (36a), and in auxiliary-initial clauses, it precedes the highest AUX (36b). It is ungrammatical in these configurations for the absolutive DP to surface in either its base position or SPEC,*voice*P.

(36) *Quantified Absolutives: Second Step*

a.

Mau	[$_{\text{ASPP}}$	<u>**inggannanna**</u>	<u>**ujiang**</u>	yari	**mo**	u-nilai	Δ],
though		all	test	COMPLETE	NOW	1ERG-grade	

‘Though now I’ve graded all the tests,’

b.

Mau	[$_{\text{ASPP}}$	<u>**ing.**</u>	<u>**sanaeke**</u>	na-baca-ngang	**mo**	guru	**ian.**	**buku**	Δ],
though		all	kid	3ERG-read-APPL	NOW	teacher	each	book	

‘Though now the teacher has read each book to all the kids,’

The absolutive DP takes the same position when it hosts these quantifiers in regular clauses that contain T^0. In that context, it precedes auxiliaries and raised verbs when absolutive agreement lowers to Σ^0 (37a) and also when absolutive agreement raises to C^0 (37b). Here as above, this shift occurs without regard to the Grouping Constraint.

(37) *Quantified Absolutives: Finite Clauses*

a.

<u>**Inggannanna**</u>	<u>**ujiang**</u>	[$_{\Sigma\text{P}}$	yari	**mi**	u-nilai	Δ].
all	test		COMPLETE	NOW.3ABS	1ERG-grade	

‘Now I’ve graded all the tests.’

b.

Pirang	**pai**	<u>**inggannanna**</u>	<u>**ujiang**</u>	[$_{\Sigma\text{P}}$	na	mu-nilai	Δ]?
when	YET.3ABS	all	test		FUTURE	2ERG-grade	

‘When will you have graded all the tests?’

3.3 Chain Reduction in Context

To capture these facts, I propose that the absolutive argument systematically raises along the path in (38a): it moves to SPEC,*voice*P and then further to SPEC,ASPP. These steps allow it to escape Condition C effects from the EXT, bind variables and induce Condition C effects in that DP, and escape the clause-internal phase to interact with T^0. To capture the typical word order, I propose that this step is preferentially covert (38b). When the absolutive DP surfaces in either of its higher positions, I propose that this preference is derailed by the properties of the prenominal quantifiers or the Grouping Constraint (38c).

(38) *Movement and Reduction*

a.	SYNTAX:	$[_{\text{TP}}$	DP_{ABS}	v	$[_{voice\text{P}}$	DP_{ABS}	DP_{ERG}	$[_{voice\text{P}}$	DP_{ABS}]]].
b.	USUAL PF:	$[_{\text{TP}}$	Δ	v	$[_{voice\text{P}}$	Δ	DP_{ERG}	$[_{voice\text{P}}$	DP_{ABS}]]].
c.	SPECIAL PF:	$[_{\text{TP}}$	DP_{ABS}	v	$[_{voice\text{P}}$	Δ	DP_{ERG}	$[_{voice\text{P}}$	Δ]]].

The ensuing analysis turns on three architectural claims that are well-established in the literature on Chain Reduction. First, it must be possible for patterns of Chain Reduction to be shaped by the idiosyncratic properties of functional heads. On the view that both overt and covert movement occur in the syntax (Bobaljik, 1995, 2002), this view follows directly from the observation that certain types of movement are systematically covert (for instance, certain steps of WH-movement: Huang 1982; Richards 1997; Nissenbaum 2000; Bošković 2002). To capture the facts at hand, I propose that the system of reduction must be guided by features of two types: features that fall on heads along the clausal spine, which demand the realization of arguments in the positions they define (39), and features that fall on heads in the extended NP, which force those constituents to be realized in the highest positions they take (40).

(39) *Preferences along the Spine*

a.	SYNTAX:	$[_{\text{FP}}$	DP	F^0	$[_{\text{GP}}$	$\text{G}^0{}_{[\text{P-EPP}]}$	DP]].
b.	PF:	$[_{\text{FP}}$	Δ		$[_{\text{GP}}$		DP]].

(40) *Preferences within the NP*

a.	SYNTAX:	$[_{\text{FP}}$	$\text{DP}_{[\text{P-EPP}]}$	F^0	$[_{\text{GP}}$	$\text{G}^0{}_{[\text{P-EPP}]}$	DP]].
b.	PF:	$[_{\text{FP}}$	DP		$[_{\text{GP}}$		Δ]].

Second, it must be possible for these head-level preferences to be derailed by PF constraints. Many types of A-movement that are usually overt, for instance, have been argued to apply covertly in certain contexts to satisfy constraints on adjacency—from object shift in Scandinavian (Bobaljik, 2002) to subject raising in Greek (Spyropoulos & Revithiadou, 2007). In the same vein, Bošković 2002 argues that WH-movement is forced to be covert in Romanian when overt movement would violate a haplology constraint: when non-subject WH-phrases are homophonous with WH-subjects, they seem to remain in place (41).

(41) *PF-Constraints Shape Chain Reduction*

a. Cine **ce** precede ~~ce~~?
who what precedes
'Who precedes what?'

b. Ce ~~ce~~ precede **ce**?
What what precedes what
'What precedes what?'

Third, it must be possible for these preferences to be derailed by phonological output constraints. In South Slavic languages, for instance, monosyllabic pronouns and auxiliaries are often realized low to satisfy constraints that ban prosodically minimal elements at the left edge of the intonational phrase Franks 1998; Bošković 2001; Harizanov 2014b. Franks & King 2000 argue that these constraints yield the second-position effect in (42): in Serbo-Croatian, clitic auxiliaries and pronouns raise through several positions in the syntax and undergo insertion in the highest position that is non-initial in the utterance.

(42) *Output Constraints Shape Chain Reduction*

a. Predstavio sam mu se.
introduced am to him REFLEXIVE
'I introduced myself to him.' Franks & King 2000 (reglossed)

b. [$_{FP}$ ~~SAM MU~~ ~~SE~~ [Predstavio sam mu se [~~PREDSTAVIO~~]]]
am to him REFLEXIVE introduced
'I introduced myself to him.' Franks & King 2000 (reglossed)

The following section aims to integrate these ingredients with the overarching theory of Chain Reduction. Starting from the interaction with output constraints, I propose that Chain Reduction must occur within the phonology. To capture this result, I propose a slight modification to the theory of Vocabulary Insertion—and from there, work toward an architectural vision of how information flows through the modules of the PF branch.

4 Chain Reduction

In Distributed Morphology, the standard theory links Chain Reduction to the system of Exponence. The terminal nodes of the morphosyntax, on this view, contain abstract bundles of features that only acquire phonological content through a postsyntactic process of Vocabulary Insertion (Embick 2015). I assume that this process proceeds in two steps. First, terminal nodes like (43a) are matched with exponents like (43b) at a stage that follows many morphological operations, like impoverishment and linearization (Halle & Marantz, 1993; Adger, 2006; Arregi & Nevins, 2012; Hewett, 2023). Second, the exponents selected in this way are then introduced in these nodes by the process of insertion in (43c) (and for further discussion, see Bye & Svenonius 2012; Kalin 2022; Hewett & Kramer 2026).

(43) *Insertion and Exponence*

a. *Feature Bundles*

x^0

[F$_1$, F$_2$, …]

b. *Matching*

x^0

[F$_1$, F$_2$, …]

[α]

c. *Insertion*

ω

[α]

Following Saab 2008, 2017 and Muñoz-Pérez 2018, I assume that Chain Reduction involves a failure of insertion: the links that are "reduced" at PF are simply not targeted for insertion. On this analysis, Chain Reduction is not a type of deletion that suppresses constituents with phonological form. Instead, it is similar to ellipsis (Chomsky, 1995; Landau, 2006): it targets phonologically empty constituents and prevents them from being exponed (Wilder, 1997; Bartos, 2000, 2001; Kornfeld & Saab, 2002; Nunes & Zocca, 2009; Arregi, 2010; Temmerman, 2012; Merchant, 2015; Abels, 2019). The result is a parallel to syntactically licensed processes of suppression, like VPE (44a) and a separation from phonological processes like left-edge deletion (44b), which deletes prosodic constituents to satisfy phonological constraints at the left edge of the intonational phrase (Weir 2014).

(44) *Chain Reduction as Non-Insertion*

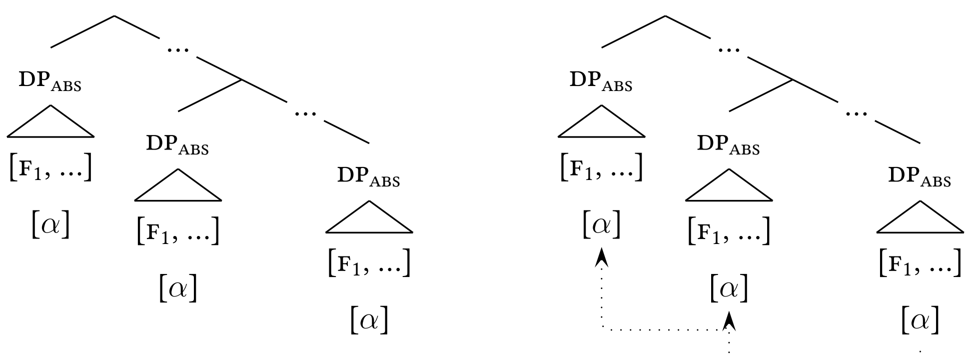
a. They got milk, and I did [$_{\text{VP}}$ Δ] too.

b. {$_\iota$ ~~have you~~ got milk? }

To reduce chains, I assume that the system of insertion operates over morphosyntactic representations in a global fashion, inspecting chains in their entirety to select a single link for insertion (Nunes, 2004). At this stage, I assume that it can access information of two types. The first are the morphosyntactic features and relationships that guide Chain Reduction, like c-command (Chomsky, 1993; Richards, 1997; Nissenbaum, 2000) and linear order (Nunes, 1995, 2004; Muñoz-Pérez, 2018). The second is the type of identity that allows constituents related by movement to be treated as "the same" for the purposes of insertion—informally speaking, to behave as links in a Chain. Following Muñoz-Pérez 2018, I assume that this identity is calculated on a featural basis in the morphological module and checked at a stage that follows morphological movements like lowering and morphological processes that render constituents invisible to Chain Reduction, like m-merger (Harizanov, 2014a). When constituents are recognized as identical in this way, on this analysis, they are grouped into a type of Chain at the stage before insertion applies.[2]

(45) *Chain Formation*

a. *Global Inspection* b. *Chain Formation*



When insertion applies, I assume that it is forced to skip every link in the chain but one in response to two constraints (see also Pesetsky 1997; Landau 2006; Bianchi 2019; Seguin 2025). I assume that these are ranked and violable constraints, in the sense of Optimality Theory (Prince & Smolensky, 2004), and propose that they are relativized versions of the pressures MAX and DEP. The first, PF-RECOVERABILITY, is a MAX constraint that demands an exponent for every chain in the morphosyntax (46a)—informally speaking, a pressure that forces insertion to occur (and see also Kurisu 2001; Henderson 2012). The second, PF-ECONOMY, is a DEP constraint that demands a unique chain in the morphosyntax for

[2]For alternative theories of Chains, Copies, and Chain-Formation, see Chomsky 1995, 2004, 2008; Nunes 2004; Martin & Uriagereka 2014; Collins & Stabler 2016; Collins & Groat 2026; on the link to Multidominance, Gärtner 1999; Kracht 2001; Frampton 2004; Gračanin-Yuksek 2007; Johnson 2007; Sauerland 2007.

every element that undergoes insertion (46b)—informally speaking, a constraint against doubling and the exponence of multiple links in a chain (and see also Nunes 2004; Jouitteau 2012; Cheng & Vicente 2013; Bjorkman 2020; Bleaman 2022). In the simplest cases, these constraints interact to ensure that insertion targets a single link in each chain.

(46) *Chain Reduction Constraints*

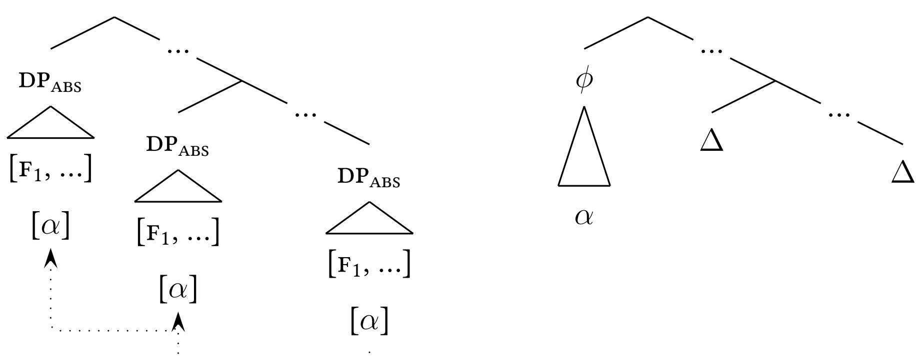
a. Pf-Recoverability
Assign one violation for every chain in the input which does not correspond to a unique exponent in the output.

b. Pf-Economy
Assign one violation for every exponent in the output which does not correspond to a unique chain in the input.

In the simplest cases, Chain Reduction favors the highest link in the chain for insertion. I assume that this preference is forced by another ranked and violable constraint (Pesetsky 1997; Bobaljik 2002; Nunes 2004). When left unchecked, it interacts with max and dep to yield the pattern in (47): insertion targets the highest link and skips the rest.

(47) *Chain Reduction*

a. *Before Insertion* b. *After Insertion*

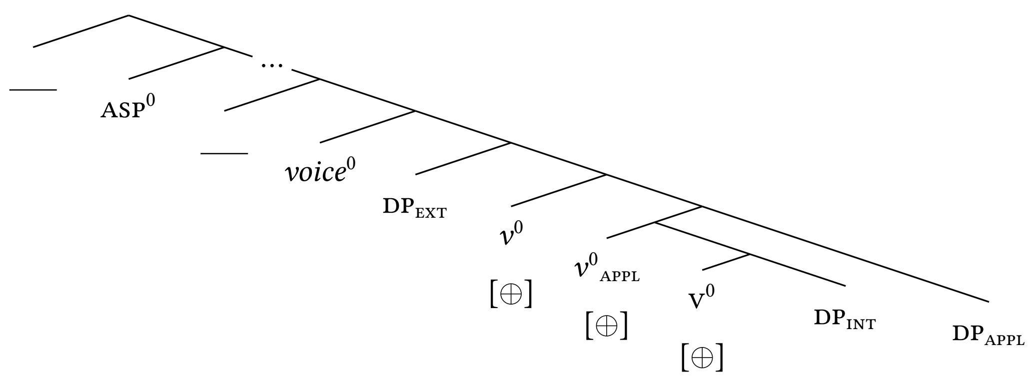


The result is a system that requires insertion to proceed in a global calculus and interact with ranked and violable constraints. To capture these properties and allow it to interact with output constraints, I propose that insertion—distinct from the morphological process that matches terminal nodes with vocabulary entries to feed it—occurs in the phonology. Following Selkirk 2009, 2011, I assume that this module takes the output representations of the morphosyntax and reworks them in a global, parallel, and constraint-based calculus that is sensitive to morphosyntactic information of many types (and see also Elfner 2012, 2015; Elordieta 2015; Bennett *et al.* 2016, 2019; Kalivoda 2018; Ishihara & Kalivoda 2022).

4.1 Lowest-Copy Spell-Out

Returning to Mandar, I propose that the preference for covert A-movement of the absolutive DP emerges from the interaction between the lexical distribution of morphosyntactic features and the language-level ranking of phonological constraints. Following Landau 2006, I assume that heads can be associated with PHONOLOGICAL EPP features that provide instructions for the system of insertion. Marking their distribution with the diacritic [⊕], I propose that these features fall on the heads v^0, *appl*^{0}, and V^0. I take them to be absent from the higher heads that consistently attract the absolutive DP: *voice*^{0} and $\text{ASP}_{\text{VIEW}}{}^0$ (48).

(48) *Phonological EPP Features*



I propose that the feature [⊕] forces insertion to target the positions selected by these heads: SPEC,*v*P, SPEC,*appl*P, and the complement position of V^0. I assume that these features are present in the syntax, preserved in the morphology, and visible in the representations that form the inputs to the phonology. In that module, I propose that their effects are imposed by an isomorphism constraint that demands the faithful execution of featural instructions from the morphosyntax. I take this to be a gradient constraint that demands insertion into the link closest to the feature [⊕] (49a). Given the distribution of [⊕] in Mandar, this will conflict with the pressure for highest-copy realization in (49b).

(49) *Positional Constraints*

a. EXECUTE

Assign one violation for every link in an input chain that separates the target of insertion from the feature [⊕].

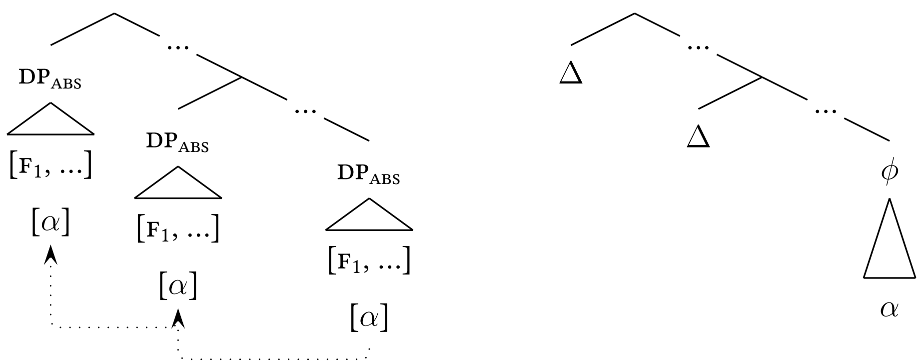
b. HIGHEST

Assign one violation for every chain in the input for which insertion does not target the highest possible link.

To force insertion to target lower positions, I propose that Execute must be ranked above Highest. In the absence of interference from other constraints, this ranking forces the pattern of Chain Reduction in (50): the absolutive DP is realized in its base position.

(50) *Low Insertion*

a. *Morphosyntax* b. *Phonology*



4.2 Interference from the Morphosyntax

To capture the effects of *inggannanna* "all" and *ianasanna* "each," I propose that these quantifiers spell out heads with phonological instructions of a second type: features that demand insertion into the highest possible position. Representing this feature with the diacritic [$\otimes$], I propose that it falls on the heads that host *inggannanna* and *ianasanna* and not the head that hosts *nasang* "every." I will refer to it as a High Spell-Out feature and assume that it percolates upwards from its associated head to the entire containing XP.

(51) *High Spell-Out Features*

a. *Prenominal Quantifiers* b. *Postnominal Quantifiers*

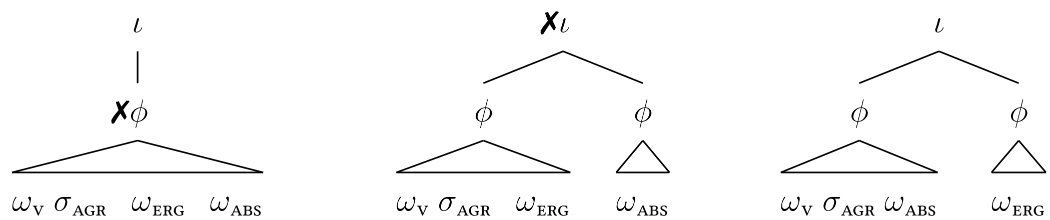
To force the absolutive DP to be realized in the highest possible position whenever it contains a quantifier of this type, I propose that [$\otimes$] is present in the syntax, preserved in the morphology, and visible in the the input to the phonology, where its effects are enforced by an analogous type of isomorphism constraint. I propose that it this second constraint is ranked above Execute to force the opposite pattern of reduction: in clauses that project up to ASP_{VIEW}P, the absolutive DPs that contain [$\otimes$] are realized in SPEC,ASPP.

5 The Prosody of Preposing

In contexts of Absolutive Preposing, I propose that the instructions of the morphosyntax are derailed by a pressure to parse absolutive agreement and the absolutive DP into a surface phonological domain. This is a constraint that mobilizes the output constituent structure of the phonology to reflect an input relationship of the morphosyntax, falling together with a wider family of pressures that I will refer to as *Interface Constraints*. Its effect on the system of Chain Reduction emerges from its relative ranking with EXECUTE and a constraint on weight: the relevant prosodic domain can only contain three words. When this pressure would separate the absolutive DP from absolutive agreement, the Grouping Constraint forces insertion of the absolutive DP into higher positions along its chain.

(52) *The Structure of Preposing*

a. *Weight* b. *Problem* c. *Resolution*



5.1 Prosodic Domains

The case for this analysis begins from the building blocks of prosodic constituency. Phonological strings are transparently parsed into a layered and hierarchical constituent structure that defines the domains for phonotactic restrictions and prosodic constraints in the surface phonology (Selkirk, 1984, 1986; Nespor & Vogel, 1986). This type of constituent structure is built from a series of abstract and universal phonological domains: the metrical categories of mora, syllable, and foot, the interface categories of prosodic word and phonological phrase, and the pragmatic category of the intonational phrase (Itô & Mester, 2007, 2009, 2012, 2013). Together, these domains form the Prosodic Hierarchy in (53).

(53) *The Prosodic Hierarchy*

μ	$>$ σ	$>$ FT	$>$ ω	$>$ ϕ	$>$ ι
MORA	SYLLABLE	FOOT	WORD	PHRASE	INTONATIONAL PHRASE

Following the standard theory of prosodic structure-building, I assume that the prosodic word and phonological phrase are built around the output correspondents of specific morphosyntactic domains to satisfy input-output mapping constraints (McCarthy & Prince, 1993; Selkirk, 1995; Truckenbrodt, 1995, 1999). I assume that phonological phrases are preferentially built around the correspondents of all syntactic phrases (Selkirk, 2009, 2011) and that prosodic words are preferentially built around the correspondents of morphological words (on which see Embick & Noyer 2001; Embick 2010, 2015; Arregi & Nevins 2012). Following Itô & Mester 2019, I take these patterns of mapping to be forced by constraints of the family MAX (see also Weber 2020a,b). The constraints I assume are given in (54).

(54) *Mapping Constraints*

a. Max-Word
Assign one violation for every morphological word in the input
that does not correspond to a prosodic word in the output.

b. Max-Phrase
Assign one violation for every morphosyntactic phrase in the input
that does not correspond to a phonological phrase in the output.

5.1.1 The Prosodic Word

In Mandar, the prosodic word can be identified from the distribution of stress. In each word, the language requires moraic trochees to be built iteratively from the right around pairs of light syllables (55a), heavy-light sequences (55b), and light-heavy sequences (55c).

(55) *The Prosodic Word*

a.	b.	c.	d.
ω	ω	ω	ω
(póle)	(lámba)	(léleŋ)	(lèp)(páŋ)
'come'	'go'	'go about'	'stop by'

The distribution of stress reveals a regular and familiar asymmetry between the exponents of lexical and functional heads (Truckenbrodt 1995; Selkirk 1995). Stress falls on the exponents of all lexical heads, like nouns, adjectives, adverbs, and verbs (56a)-(56b). It also falls on certain affixes in an item-specific way, like the reduplicant (56c) and the antipassive prefix (56d). I propose that every constituent that bears stress forms a word and take this parse to follow from two constraints: the pressure to build prosodic words around morphological words (which are assembled around the lexical heads in the morphology) and the prosodic subcategorization frames of individual exponents in the lexicon (Inkelas, 1989; Raffelsiefen, 1999; Paster, 2006; Bennett *et al.*, 2018; Kalin & Rolle, 2023).

(56) *Prosodic Words*

a. [$_\omega$ bállo] [$_\omega$ **mányang**]
palm wine strong
'strong palm wine'

b. [$_\omega$ **máne**] [$_\omega$ na-rúndu]
just 3ERG-drink
'just drank'

c. [$_\omega$ **náru**] [$_\omega$ na-rúndu]
RED 3ERG-drink
'sorta drink'

d. [$_\omega$ **mán**] [$_\omega$ dúndu]
ANTIP drink
'drink'

Stress never falls on functional elements like prepositions and demonstratives (57a). In the usual case, it is also absent from auxiliaries (57b). When these elements lack stress, I propose that they do not form words and simply fall into the words built around their associated lexical heads (Peperkamp, 1997; Booij, 1996; Vigário, 1999; Itô & Mester, 2009).

(57) *Non-Words*

a. [$_\omega$ **pole ro** pónna] [$_\omega$ ó]
from that tree there
'From that tree there.'

b. [$_\omega$ mo **ndang** na-rúndu],
though not 3ERG-drink
'Though he didn't have it,'

5.1.2 The Phonological Phrase

The phonological phrase can be identified from the distribution of a high tone that falls at its right edge. This tone falls after strings of functional and lexical heads (58a), lexical heads and head-level adjuncts (58b), norman genitives (58c), and complex names (58d).

(58) *The Phonological Phrase*

a. {$_\phi$ [$_\omega$ tama ri wóyang] }$^{\text{H}}$
into P^0 house
'Into the house'

b. {$_\phi$ [$_\omega$ ánde] [$_\omega$ maráras] }$^{\text{H}}$
food spicy
'Spicy food'

c. {$_\phi$ [$_\omega$ bírit] [$_\omega$ tàngngalálang] }$^{\text{H}}$
edge road
'Edge of the road'

d. {$_\phi$ [$_\omega$ júpe] [$_\omega$ táli'] }$^{\text{H}}$
NAME NAME
'Jupri Talib'

The same tone also falls after the auxiliaries that host absolutive agreement and viewpoint aspect (59a)-(59b). In this context, the hosting auxiliary acquires stress as well. I propose that this shift is forced by the lexical requirements of the absolutive and aspectual clitics in a second type of prosodic subcategorization: the highest auxiliary must form a word and a phrase to satisfy the output requirements of the exponents of T^0 and $\text{ASP}_{\text{VIEW}}{}^0$.

(59) *The Prosodic Promotion Effect*

a. **yári**ᴴ **mi** lámbaᴴ.
COMPLETE NOW.3ABS go
'Now they've gone.'

b. mo **yári**ᴴ **mo** lámbaᴴ,
though COMPLETE NOW go
'Though now they've gone,'

5.1.3 The Maximal Phonological Phrase

The distribution of stress and the high tone reveals that prosodic words and phonological phrases are built around four types of elements in regular clauses: the verbal complex, phrasal arguments, phrasal adjuncts, and the auxiliaries that host the absolutive and aspectual clitics. Phonological phrases are also built around many functional projections along the clausal spine, yielding the type of prosodic recursion that characterizes the phonological phrase (Itô & Mester 2007, 2012, 2013; Itô & Mester 2021; Elfner 2012, 2015; Elordieta 2015; Van Handel 2021; also Ladd 1986; Wagner 2005, 2010). Example (60) illustrates this parse in a five-line format that I will use to present examples below: the top line shows the highest visible level of phonological phrasing, the second line shows the surface form of each constituent, and the third line shows underlying forms.

(60) *Recursive Phonological Phrasing*

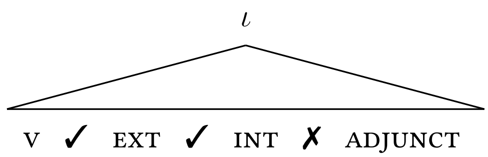
$\{_{\phi[\text{MAX}]}$			}	$\{_{\phi[\text{MAX}]}$ }
néteᴴ	pe	ɣúruᴴ	ɹótorᴴ	**dío**ᴴ
na-itai	pai	guru	dottor	dio
3ERG-look for	YET.3ABS	teacher	doctor	there

'The teacher will look for the doctor there.'

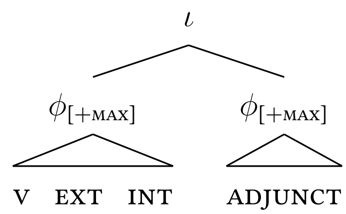
The evidence for this higher level of phonological phrasing emerges from a series of restrictions on segmental processes in the phrasal phonology. These restrictions emerge under the neutral phonological conditions that facilitate the emergence of unmarked patterns of phonological phrasing (Nespor & Vogel, 1986): in well-planned and broad-focus utterances produced at a regular rate of speech. In this context, speakers are able to provide the categorical judgment in (61a): these processes preferentially apply between most prosodic words and phonological phrases, but they are blocked at specific junctures within the intonational phrase. Brodkin 2025a proposes that these restrictions are keyed to the intermediate level of prosodic structure in (61b): these processes are blocked at the left and right edges of the MAXIMAL PHONOLOGICAL PHRASE, which is the type of phonological phrase directly dominated by the intonational phrase (Itô & Mester, 2007, 2012, 2013).

(61) *The Maximal Phonological Phrase*

a. *Restrictions*



b. *Constituency*



The following discussion will illustrate the distribution of maximal phonological phrases with four processes of this type (Brodkin 2025a; for a fifth, see Brodkin 2025b). The first three are blocked at its right edge: a process of Falling-Sonority Coalescence that reduces sequences like /*ai*/ to [*e*], a process of Rising-Sonority Reduction that reduces sequences like /*ia*/ to [*e*], and a process of Coda Deletion that deletes intervocalic /ʔ ŋ/. The fourth is blocked at its left edge: a process of Voiced Obstruent Lenition that reduces intervocalic /*b d d͡ʒ g*/ to [*w ɹ j ɣ*]. These processes are summarized in table (62).

(62) *Diagnostics for the Maximal Phrase*

CONTEXT	PROCESS	TARGET	EFFECT
WORD-FINAL	Falling-Sonority Coalescence	/*ai ae*/ /*ao au*/ /*ei oi*/	[*e*] [*o*] [*e, o*]
	Rising-Sonority Reduction	/*ie, ue*/ /*io, uo*/ /*iu, ui*/ /*ia, ua*/	[*e*] [*o*] [*u, i*] [*e, o*]
	Intervocalic Coda Deletion	/*ʔ*/ /*ŋ*/	[Ø] [Ø + nasalization]
WORD-INITIAL	Intervocalic Lenition	/*b d g*/ /*d͡ʒ*/	[*w ɹ ɣ*] [*j*]

5.2 Phrasing Constraints

Maximal phonological phrases are always built around three-word strings of verbs and arguments in the *v*P. We can see this fact from a restriction on Voiced Obstruent Lenition. In strings of the order V-EXT-INT-ADJUNCT, lenition applies at the left edges of the EXT and the INT (63a). In strings of the order V-INT-GOAL-ADJUNCT-ADJUNCT, it applies at the left edges of the INT and the GOAL (63b). It is blocked at the edge of every following adjunct.

(63) *Maximal Phonological Phrasing*

a.

{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }
néte^{H}	pe	**ɣ**úru^{H}	**ɹ**ótor^{H}	**d**ío^{H}
na-itai	pai	guru	<u>dottor</u>	dio
3ERG-look for	YET.3ABS	teacher	doctor	there

'The teacher will look for the doctor there.'

b.

{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }	{$_{\phi[\text{MAX}]}$ }
nállem^{H}	mi	**w**úku^{H}	**ɣ**úru^{H}	**d**ío^{H}	**d**íteʔe^{H}
na-alli-aŋ	mi	buku	<u>guru</u>	dio	diteʔe
3ERG-buy-APPL	NOW.3ABS	book	teacher	there	now

'Now they're buying books for the teacher there.'

Maximal phonological phrases are also built around three-word strings of the orders AUXILIARY-VERB-ARGUMENT (64a) and WH-VERB-ARGUMENT (64b). In these strings, lenition applies at the left edges of the V and the DP but not in final adjuncts to the *voice*P.

(64) *Maximal Phonological Phrasing: Auxiliaries*

a.

{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }	{$_{\phi[\text{MAX}]}$ }
jári^{H}	pe	**ɹ**íta^{H}	**ɣ**úru^{H}	**d**ío^{H}	**d**éʔ^{H}
iari	pai	di-ita	<u>guru</u>	dio	de
COMPLETE	YET.3ABS	2ERG-see	teacher	there	PRT

'You'll see the teacher there, eh?'

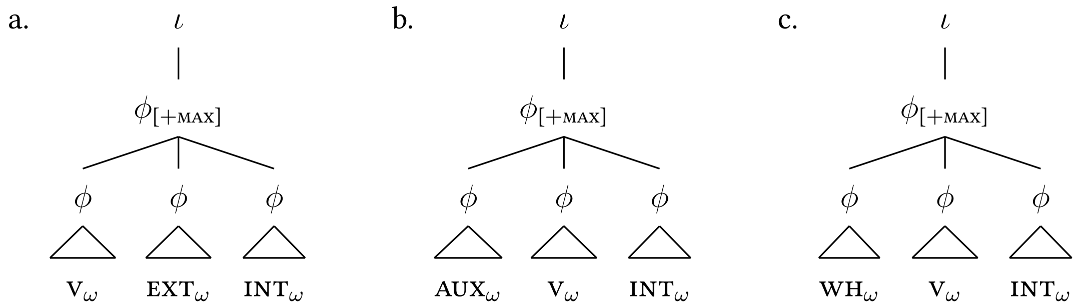
b.

{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }
pírap^{H}	pe	**ɣ**umóra^{H}	**ɣ**úru	**d**ío^{H}
piraŋ	pai	gumora	<u>guru</u>	dio
when	YET.3ABS	scream	teacher	there

'When will the teacher will scream there?'

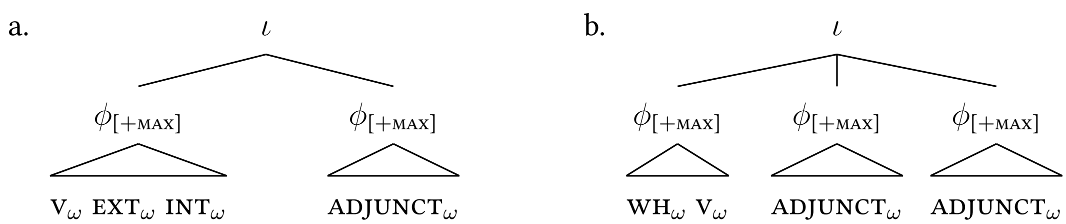
This effect reveals that phonological phrases are always built around three-word strings that follow the order WH-AUX-VERB-EXT-INT-GOAL. These phrases correspond to functional projections along the clausal spine and they are built to satisfy the pressure to map every input XP to an output ϕ (65). For ease of exposition, I set aside the questions of exactly how many functional projections are typically mapped to phrases in this way.

(65) *Maximal Phonological Phrasing*



Every *voice*P-level adjunct forms its own maximal phonological phrase (66). This parse traces a stable cross-linguistic fact: adjuncts to the extended VP are always excluded from the phonological phrases built around the projections of the clausal spine (Brodkin 2025a; see also Hale & Selkirk 1987; Chen 1987; Truckenbrodt 1995, 1999, Selkirk 2011, p.483, fn.38; for discussion of Truckenbrodt 1995's facts in different terms, Zubizarreta 1998).

(66) *Parsing Adjuncts*



5.2.1 The Weight Constraint

When we add additional words to the WH-AUX-VERB-ARGUMENT string, we see different patterns of phonological phrasing. When individual constituents contain three prosodic words, they never undergo left-edge lenition: for instance, in the V-EXT-INT string in (67a), where the INT contains three words. When constituents contain two words, in turn, they never undergo lenition when they follow constituents that also contain two words: for instance, in the V-EXT-INT string in (67b), where each constituent contains two words.

(67) *The Weight Constraint*

a.

{$_{\phi[\text{MAX}]}$		}	{$_{\phi[\text{MAX}]}$		}	{$_{\phi[\text{MAX}]}$ }
nete$^{\text{H}}$	pe	ɹótor$^{\text{H}}$	**g**úru	ílmu	wása$^{\text{H}}$	**d**ío$^{\text{H}}$
na-itai	pai	dottor	guru	ilmu	basa	dio
3ERG-look for	YET.3ABS	doctor	teacher	science	language	there

'The doctor will look for the linguistics teacher there.'

b.

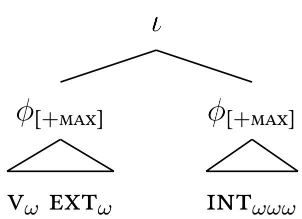
{$_{\phi[\text{MAX}]}$	}	{$_{\phi[\text{MAX}]}$	}	{$_{\phi[\text{MAX}]}$	}	{$_{\phi[\text{MAX}]}$ }
néte-néte$^{\text{H}}$	pai	**d**ótor	ríŋe$^{\text{H}}$	**g**úru	wása$^{\text{H}}$	**d**ío$^{\text{H}}$
($\sigma\sigma$)-na-itai	pai	dottor	riŋe	guru	basa	dio
RED-3ERG-look for	YET.3ABS	doctor	tooth	teacher	language	there

'The dentist will look for the language teacher there.'

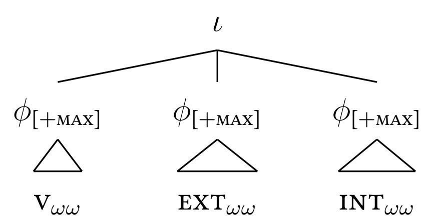
This shift reveals a constraint on weight (Nespor & Vogel, 1986; Kubozono, 1989): in Mandar, the maximal phonological phrase cannot contain more than three prosodic words. I will refer to it as the MAXIMAL TERNARITY CONSTRAINT. When lenition is blocked in the final argument in the *voice*P, I propose that it triggers the shift in (68a): the final argument is excluded from the phonological phrases built around projections on the clausal spine and it comes to be directly dominated by the intonational phrase on its own. When lenition is blocked in every XP in the *voice*P, I propose that it drives the shift in (68b).

(68) *Maximal Ternarity*

a. *Final Arguments*



b. *All Arguments*



5.2.2 The Isomorphism Constraint

When additional prosodic words are added to the verbal complex, it is often prevented from forming a maximal phonological phrase with every argument in the *voice*P. In that context, non-absolutive arguments in the *voice*P are typically parsed into maximal phonological phrases of their own. When the verb is antipassive, for instance, we can see this effect from a pattern of allomorphy laid out in Brodkin 2025a. When the verbal root begins with a vowel, the antipassive prefix typically takes the form *m-* and V-INT-GOAL strings are typically parsed into maximal phonological phrases (69a). When the root begins with a consonant, the prefix takes the form *maN-* and forms an independent prosodic word. In that context, the verbal complex typically forms a maximal phonological phrase and the INT-GOAL string forms a maximal phonological phrase of its own (69b).

(69) *Antipassive Allomorphy: Prosodic Asymmetry*

a.

{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }
maláŋap^{H}	pe	**w**úku^{H}	**ɣ**úru^{H}	**d**ío^{H}
maʔ-ala-ŋaŋ	pai	buku	guru	dio
ANTIP-get-APPL	YET.3ABS	book	teacher	there

'They'll get teachers books there.'

b.

{$_{\phi[\text{MAX}]}$	}	{$_{\phi[\text{MAX}]}$	}	{$_{\phi[\text{MAX}]}$ }
mámbéŋap^{H}	pai	**b**úku^{H}	**ɣ**úru^{H}	**d**ío^{H}
maŋ-be-ŋaŋ	pai	buku	guru	dio
ANTIP-give-APPL	YET.3ABS	book	teacher	there

'They'll give teachers books there.'

When the verb is ditransitive, we see the same effect in strings of the order V-EXT-INT. When the verbal complex is one word long, the full string typically forms a maximal phonological phrase (70a). When the verbal complex contains two, the verbal complex and the EXT-INT string are parsed into separate maximal phonological phrases (70b).

(70) *Postverbal Constituents*

a.

{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }
nállep^{H}	pe	**ɣ**úru^{H}	**w**úku^{H}	**d**ío^{H}
na-alli-aŋ	pai	guru	buku	dio
3ERG-buy-APPL	YET.3ABS	teacher	book	there

'The teacher will buy them books there.'

b.

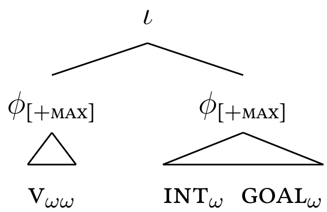
{$_{\phi[\text{MAX}]}$			}	{$_{\phi[\text{MAX}]}$ }	{$_{\phi[\text{MAX}]}$ }
jári^{H}	pe	náll**eŋ**	gúru	**b**úku^{H}	**d**ío^{H}
jari	pai	na-alli-aŋ	guru	buku	dio
COMPLETE	YET.3ABS	3ERG-buy-APPL	teacher	book	there

'The teacher will buy them books there.'

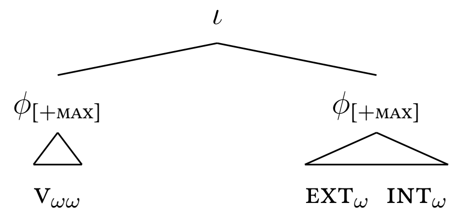
Brodkin 2025a argues that these postverbal phonological phrases correspond to headless functional projections in the extended VP. When the verbal complex contains two words, on this analysis, Maximal Ternarity interacts with the pressure to map input XPs to output ϕs to yield the parses in (71): constituents like the *v*P are mapped to phonological phrases, they cannot be parsed into the larger phonological phrase built around the *voice*P, and as result, they are revealed by our tests for maximal phonological phrasing.

(71) *Match Constraints*

a. *Antipassives*



b. *Ditransitives*



5.2.3 The Attraction Constraint

When absolutive agreement follows auxiliaries and WH-phrases outside the *voice*P, we see a different effect. In four-word strings that follow the order WH-AUX-V-EXT-INT-GOAL, the verbal complex is preferentially parsed together with preceding auxiliaries and WH-phrases over the arguments in the *v*P. The following examples illustrate this fact with High-Vowel Reduction in strings that contain the antipassive verb *malliaŋ* "buy for." In three-word V-INT-GOAL strings, this verb takes the form [*malleŋ*] and forms a maximal phonological phrase with the INT and GOAL in the *v*P (72a). In four-word strings of the orders WH-V-INT-GOAL and AUX-V-INT-GOAL, it takes the form [*malliaŋ*] (72b)-(72c).

(72) *Syntax-Prosody Mismatch*

a.

$\{_{\phi[\text{MAX}]}$			}	$\{_{\phi[\text{MAX}]}$ }
mállep$^{\text{H}}$	pe	**wú**ku$^{\text{H}}$	**ɣú**ru$^{\text{H}}$	**dí**o$^{\text{H}}$
maʔ-alli-aŋ	pai	buku	guru	dio
ANTIP-buy-APPL	YET.3ABS	book	teacher	there

'They'll buy teachers books there.'

b.

$\{_{\phi[\text{MAX}]}$		}	$\{_{\phi[\text{MAX}]}$	}	$\{_{\phi[\text{MAX}]}$ }
jári$^{\text{H}}$	pe	mall**íam**	búku$^{\text{H}}$	**ɣ**úru	**dí**o$^{\text{H}}$
jari	pai	maʔ-alli-aŋ	buku	guru	dio
COMPLETE	YET.3ABS	ANTIP-buy-APPL	book	teacher	there

'They'll buy teachers books there.'

c.

$\{_{\phi[\text{MAX}]}$		}	$\{_{\phi[\text{MAX}]}$	}	$\{_{\phi[\text{MAX}]}$ }
pírap$^{\text{H}}$	pe	mall**íam**	búku$^{\text{H}}$	**ɣ**úru	**dí**o$^{\text{H}}$
piraŋ	pai	maʔ-alli-aŋ	buku	guru	dio
when	YET.3ABS	ANTIP-buy-APPL	book	teacher	there

'When will they buy teachers books there?'

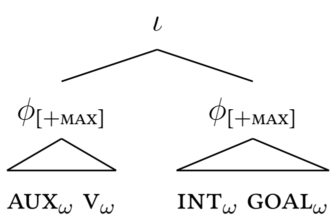
I propose that this parse is forced by a pressure to group the verb with functional elements in the extended TP, like auxiliaries and the absolutive and aspectual clitics. This pressure belongs to a family of interface constraints that mobilize surface phonological constituency to reflect the relationships that hold between functional and lexical heads in the morphosyntax. I propose that it is a ranked and violable pressure with the force in (73a), analogous to the constraints that force many other types of functional elements to be parsed into larger prosodic words with the exponents of their associated lexical heads (Peperkamp 1997; Booij 1996; Vigário 1999; Itô & Mester 2009; see also Selkirk 1995; Hall 1999; Werle 2009). In the contexts above, it forces the parse in (73b), where the verb is parsed together with auxiliaries and the hosts of absolutive agreement and viewpoint aspect, and it rules out the parse in (73c), where the verb is parsed with postverbal arguments into maximal phonological phrases that straightforwardly correspond to the *voice*P.

(73) *The Attraction Constraint*

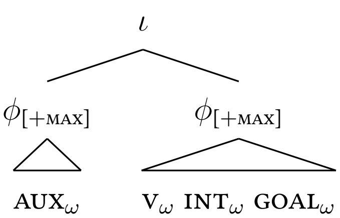
a. ATTRACT

Assign one violation for every functional element that is not parsed into the same phonological phrase as the exponent of its corresponding lexical head.

b. *Attraction*



c. *Isomorphism*



5.3 Absolutive Preposing Revised

The absolutive DP surfaces in its base position whenever the usual rules of mapping should parse an argument in that position in the same maximal phonological phrase as the host of absolutive agreement. In verb-initial clauses, as a result, it takes its usual position in the EXT-INT-GOAL string when that position falls into a three-word window with the verb.

(74) *Grouping from the Base Position*

a. $\{_{\phi[\text{MAX}]}$ $\}$ $\{_{\phi[\text{MAX}]}$ $\}$

néte$^{\text{H}}$	pe	**ɣ**úru$^{\text{H}}$	**ɹ**ótor$^{\text{H}}$	**dío**$^{\text{H}}$
na-itai	pai	guru	<u>dottor</u>	dio
3ERG-look for	YET.3ABS	teacher	doctor	there

'The teacher will look for the doctor there.'

b. {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ }

nállem$^{\text{H}}$	mi	**wúku**$^{\text{H}}$	**ɣúru**$^{\text{H}}$	**dío**$^{\text{H}}$	**díteʔe**$^{\text{H}}$
na-alli-aŋ	mi	buku	guru	dio	diteʔe
3ERG-buy-APPL	NOW.3ABS	book	teacher	there	now

'Now they're buying books for the teacher there.'

When absolute agreement follows auxiliaries and WH-phrases, we see the same effect: the absolutive DP is realized in its base position whenever it should fall into the same maximal phonological phrase as the host of absolutive agreement from there (75a)-(75b).

(75) *Extending the Paradigm*

a. {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ }

jári$^{\text{H}}$	pe	ɹíta$^{\text{H}}$	**ɣúru**$^{\text{H}}$	**dío**$^{\text{H}}$	**déʔ**$^{\text{H}}$
iari	pai	di-ita	guru	dio	de
COMPLETE	YET.3ABS	2ERG-see	teacher	there	PRT

'You'll see the teacher there, eh?'

b. {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ }

pírap$^{\text{H}}$	pe	**wémme**$^{\text{H}}$	**wúa**$^{\text{H}}$	**dío**$^{\text{H}}$	**díteʔe**$^{\text{H}}$
piraŋ	pai	bemmeʔ	bua	dio	diteʔe
when	YET.3ABS	fall	fruit	there	now

'When will the fruit fall there now?'

It also takes its usual position in the EXT-INT-GOAL string when the collective distribution of words makes it impossible to satisfy the Grouping Constraint. This effect is forced whenever the absolutive DP or the host of agreement individually contains three words, as in (76a), or whenever the two contain more than three words collectively, as in (76b).

(76) *Giving up on Grouping*

a. {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ }

nete$^{\text{H}}$	pe	ɹótor$^{\text{H}}$	**gúru**$^{\text{H}}$	ílmu	wása	**dío**$^{\text{H}}$
na-itai	pai	dottor	guru	ilmu	basa	dio
3ERG-look for	YET.3ABS	doctor	teacher	science	language	there

'The doctor will look for the linguistics teacher there.'

b. {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ } {$_{\phi\text{[MAX]}}$ }

jás	**sápa**$^{\text{H}}$	pe	netái	**gúru**	**wása**$^{\text{H}}$	**dío**$^{\text{H}}$
jaŋ	saʔapa	pai	na-itai	guru	basa	dio
time	what	YET.3ABS	3ERG-look for	teacher	language	there

'What time will they look for the language teacher there?'

5.3.1 Higher Spell-Out

When the absolutive DP cannot satisfy the Grouping Constraint from its base position, it is realized in SPEC,*voice*P whenever the usual mapping would parse it into a maximal phonological phrase with the host of agreement from there. In verb-initial strings where the he verb is transitive, this constraint forces realization in SPEC,*voice*P whenever the INT and the verbal complex collectively contain three words (77a)-(77b). When the INT and the verbal complex collectively contain two words, we see the same effect when the base position of the INT should be separated from the verb by a two-word EXT (77c).

(77) *Absolutive Preposing*

a. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

nete-nete[H]	pe	**ɣúru**[H]	**dótor**[H]		**dío**[H]
($\sigma\sigma$)-na-itai	pai	<u>guru</u>	dottor	Δ	dio
RED-3ERG-look for	YET.3ABS	teacher	doctor		there

'The doctor will look around for the teacher there.'

b. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

nete[H]	pe	**ɣúru**[H]	wása	**dótor**[H]		**dío**[H]
na-itai	pai	<u>guru</u>	<u>basa</u>	dottor	Δ	dio
3ERG-look for	YET.3ABS	teacher	language	doctor		there

'The doctor will look for the language teacher there.'

c. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

nete[H]	pe	**ɣúru**[H]	**dótor**	sanaéke[H]		**dío**[H]
na-itai	pai	<u>guru</u>	dottor	sanaeke	Δ	dio
3ERG-look for	YET.3ABS	teacher	doctor	children		there

'The pediatrician will look for the teacher there.'

When absolutive agreement follows auxiliaries, the absolutive DP surfaces in SPEC,*voice*P when the absolutive DP and the auxiliary collectively contain three prosodic words (78b). When they collectively contain two words, the absolutive DP is also realized in SPEC,*voice*P when its base position should be separated from the auxiliary by a two-word verb (78c).

(78) *The First Step*

a. {ϕ[MAX] } {ϕ[MAX] }

jári[H]	pe	**ɣumóra**[H]	**ɣúru**	**dío**[H]
jari	pai	gumora	<u>guru</u>	dio
COMPLETE	YET.3ABS	scream	teacher	there

'The teacher will scream there.'

b. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

jári[H]	pe	ɣúru	wása	gumóra[H]		**dío**[H]
jari	pai	guru	basa	gumora	Δ	dio
COMPLETE	YET.3ABS	teacher	language	scream		there

'The language teacher will scream there.'

c. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

jári[H]	pe	ɣúru	gúmo-ɣumóra[H]		**dío**[H]
jari	pai	guru	($\sigma\sigma$)-gumora	Δ	dio
COMPLETE	YET.3ABS	teacher	RED-scream		there

'The teacher will scream a bit there.'

The absolutive DP is also realized in SPEC,*voice*P where its base position should be split from the host of agreement by a one-word verb and a one-word argument in the *voice*P when absolutive agreement follows auxiliaries (79a) and WH-phrases in SPEC,CP (79b).

(79) *Parallel Separations*

a. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

jári[H]	pe	ɣúru	néte	**dótor**[H]		**dío**[H]
jari	pai	guru	na-itai	dottor	Δ	dio
COMPLETE	YET.3ABS	teacher	3ERG-look for	doctor		there

'The doctor will look for the teacher there.'

b. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

pírap[H]	pe	néte	ɣúru	**dótor**[H]		**dío**[H]
piraŋ	pai	na-itai	guru	dottor	Δ	dio
when	YET.3ABS	3ERG-look for	teacher	doctor		there

'When will the doctor look for the teacher there?'

When absolutive agreement follows WH-phrases, the absolutive DP surfaces in SPEC,ASPP whenever it can only be parsed into the same maximal phonological phrase as the host of agreement from there. This effect emerges when the absolutive DP and the WH-phrase collectively contain three words, as in (80a)-(80b). It also emerges when they collectively contain two words and the WH-phrase is split from SPEC,*voice*P by a two-word verb (80c).

(80) *The Second Step*

a. {ϕ[MAX] } {ϕ[MAX] } {ϕ[MAX] }

jás	**sápa**[H]	pe	ɣúru[H]	néte	ɹótor[H]		**dío**[H]
jaŋ	saʔapa	pai	guru	na-itai	dottor	Δ	dio
time	what	YET.3ABS	teacher	3ERG-look for	doctor		there

'What time will the doctor look for the teacher there?'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

pírap^{H}	pe	**ɣ**úru	wása^{H}	néte	**ɹ**ótor^{H}		**dío**^{H}
piraŋ	pai	guru	basa	na-itai	dottor	Δ	dio
when	YET.3ABS	teacher	language	3ERG-look for	doctor		there

'When will the doctor look for the language teacher there?'

c. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

pírap^{H}	pe	**ɣ**úru	**g**úmo-ɣumóra^{H}		**dío**^{H}
piraŋ	pai	guru	($\sigma\sigma$)-gumora	Δ	dio
when	YET.3ABS	teacher	RED-scream		there

'When will the teacher scream a bit there?'

5.3.2 The Grouping Constraint

I propose that these effects are forced by the pressure in (81): a ranked and violable interface constraint which demands that the absolutive DP fall into the same maximal phonological phrase as the corresponding exponents of absolutive agreement. This pressure belongs to a family of constraints that mobilize prosodic constituency to reflect input relationships of agreement which are most familiar from Japanese, where they reroute patterns of prosodic structure-building to group in-situ WH-phrases with interrogative c^0s (Deguchi & Kitagawa 2002; Ishihara 2003; Hirotani 2005; Smith 2005; for empirical discussion, Smith 2013, 2014; Branan 2018; for a theoretical paradigm organized around different architectural assumptions but grounded in this effect, see also Richards 2016).[3]

[3] There is one context where this constraint seems to go unenforced: when the base position of the absolutive DP is linearly adjacent to absolutive agreement, the Grouping Constraint is violated to build phonological phrases around functional projections in the *voice*P. This effect emerges when agreement follows the verb and the absolutive DP is followed by other arguments in the *voice*P, which is only possible when the verb is antipassive and the absolutive DP is leftmost in the EXT-INT-GOAL string. When the verbal complex contains two words in this context, the EXT is preferentially grouped with a following INT or GOAL when the two are collectively two words long. I leave this interaction with adjacency to future work.

(1) *Linear Adjacency: Grouping Suspended*

a. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

mámbéŋap^{H}	pai	**g**úru^{H}	**w**úku^{H}	**d**ío^{H}
maŋ-be-ŋaŋ	pai	guru	buku	dio
ANTIP-give-APPL	YET.3ABS	teacher	book	there

'The teachers will give out books there.'

b. {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ } {$_{\phi[\text{MAX}]}$ }

mámbéŋap^{H}	pai	**g**úru^{H}	**ɹ**ótor^{H}	**d**ío^{H}
maŋ-be-ŋaŋ	pai	guru	dottor	dio
ANTIP-give-APPL	YET.3ABS	teacher	doctor	there

'The teachers will give things to doctors there.'

(81) *The Grouping Constraint*

Assign one violation for every absolutive DP and corresponding exponent of absolutive agreement that are not parsed into the same maximal phonological phrase.

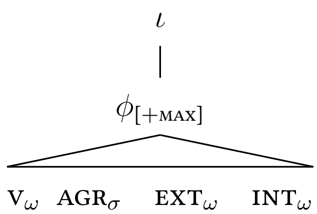
When the morphosyntax produces representations like (82a), on this analysis, the phonology turns to the Grouping Constraint to determine whether it is possible to follow instructions and insert lexical content into the lowest copy of the absolutive DP (82b). This possibility emerges when the absolutive DP can satisfy the Grouping Constraint from that position (82c) and when it cannot satisfy the Grouping Constraint at all (82d).

(82) *Low Spell-Out*

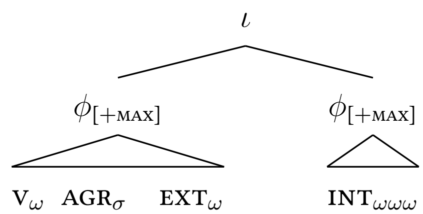
a. SYNTAX: $[_{TP}$ DP_{ABS} V $[_{voiceP}$ DP_{ABS} DP_{ERG} $[_{voiceP}$ DP_{ABS}]]].

b. FAITHFUL PF: $[_{TP}$ Δ V $[_{voiceP}$ Δ DP_{ERG} $[_{voiceP}$ DP_{ABS}]]].

c. *Low Grouping*



d. *No Grouping*



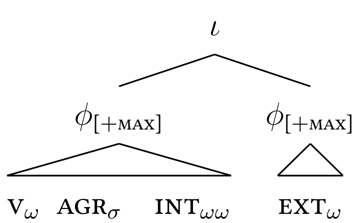
The phonology forces insertion into higher positions when the Grouping Constraint can only be satisfied from higher positions along this chain. When the absolutive DP and the host of agreement collectively form two or three prosodic words, the absolutive DP is inserted in SPEC,*voice*P whenever it can satisfy the Grouping Constraint from there and not from its base position (83a). In the same way, the absolutive DP is inserted in SPEC,*voice*P when it can satisfy the Grouping Constraint from there and not from SPEC,*voice*P (83b).

(83) *Higher Spell-Out*

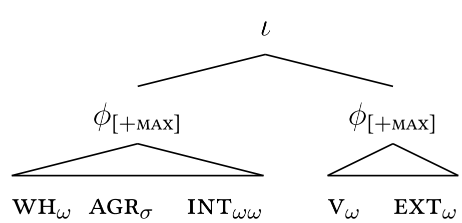
a. INTERMEDIATE: $[_{TP}$ Δ V $[_{voiceP}$ DP_{ABS} DP_{ERG} $[_{voiceP}$ Δ]]].

b. HIGHEST: $[_{TP}$ DP_{ABS} V $[_{voiceP}$ Δ DP_{ERG} $[_{voiceP}$ Δ]]].

c. *Intermediate Grouping*



d. *Higher Grouping*



The immediate result is an argument that PF-constraints can determine where insertion proceeds along the Chains produced by movement. The facts of Absolutive Preposing, in this respect, form a mirror image to the wide range of cases where PF constraints derail the preference to insert lexical content into the highest position in a chain. Drawn together with those cases, they allow us to lay out the typology of Link Selection in (84).

(84) *PF-Determined Link Selection*

Default	Derailed by	Example	Language
Highest	Anti-Homophony	Wh-movement	Romanian; Serbo-Croatian (Bošković, 2002)
	Adjacency	Object Shift	Mainland Scandinavian (Bobaljik, 2002)
		Subject Movement	Greek (Spyropoulos & Revithiadou, 2007)
	Alignment with Nuclear Stress	Subject Movement	Serbo-Crotian (Stjepanovic, 1999)
			Spanish (Ortega-Santos, 2006)
		Focus-movement	Italian (Bianchi, 2019)
		Wh-movement	Valdôtain Patois (Seguin, 2025)
Lowest	Grouping	Absolutive Preposing	Mandar

In wider perspective, these facts also reinforce the hypothesis that Chain Reduction must occur in a calculus that can access global prosodic context and interact with phonological well-formedness constraints that shape the construction of prosodic domains. Drawn together with those cases, the facts of Absolutive Preposing fall into the typology in (85).

(85) *Prosodically-Conditioned Link Selection*

Domain	Derailed by	Example	Language
Intonational Phrase	Strong Start	Second-Position Pronouns/Auxiliaries	Serbo-Croatian (Bošković, 2001) (Franks, 1998)
		Verb-Adjacent Pronouns/Auxiliaries	Bulgarian, Macedonian (Franks, 1998) (Harizanov, 2014a)
Phonological Phrase	Grouping	Absolutive Preposing	Mandar

Beyond the system of Copy Selection, these results reinforce an additional hypothesis that has emerged from work on patterns of Verb Doubling: PF-constraints can also force moved elements to undergo insertion in multiple positions along their chains (in a manner separate from the configurations where morphological operations render elements invisible to reduction: Nunes 2004; Nunes & de Quadros 2006; Corver & Nunes 2007; Cheng 2007). These effects have been argued to emerge from the analogous series of pressures in (86), which follow naturally on the theory that insertion proceeds in the global and output-oriented phonological calculus where prosodic structure is built.

(86) *PF-Conditioned Doubling Phenomena*

Trigger	Language	Source
Subcategorizational Requirements (Stray affix filter effects)	Hebrew Breton, Ingush	Landau 2006 Bjorkman 2020
Prosodic Well-Formedness	Asante Twi	Kandybowicz 2015

In broadest perspective, at last, they converge with the many lines of evidence adduced to suggest that phonological output constraints can guide the wider system of exponence. The closest parallel to Chain Reduction emerges in the domain of ellipsis, where Bennett *et al.* 2019 argue that output constraints can force lexical content to be inserted into terminal nodes marked for ellipsis in the morphosyntax. In the domain of suspended exponence, Henderson 2012 and Aissen 2017 argue for the opposite effect: output constraints can prevent exponents from being inserted into specific terminal nodes. In the domain of allomorphy, at last, the literature has advanced a broad range of arguments for an analogous result: under the right conditions, phonological output constraints must be able to guide the selection of exponents inserted into the terminal nodes of the morphosyntax. Some of the constraints that drive these alternations are listed below (and for discussion, see also Wolf 2007, 2008; Nevins 2011; Svenonius 2013; Deal & Wolf 2017).

(87) *PF-Conditioned Allomorph Selection*

Constraints	Sources
Foot Structure	Mester 1994; Drachman *et al.* 1995; Kager 1996 Dolbey 1997; Booij 1998; González 2005; Bennett 2017
Hiatus Resolution	Tranel 1996; Perlmutter 1998; Klein 2003 Bonet *et al.* 2007; Mascaró 2007; Suh 2008
Top-Down Constraints	McCarthy 2012; Brodkin 2025b

6 Conclusion

Beyond the narrow arena of Chain Reduction, the arc of our investigation leads us toward three additional results. The first surrounds the analysis of Mandar syntax in (88), which draws the absolutive DP to the edge of the clause-internal phase and then up into the middle field. The linear evidence for these two steps—drawn from the system of Absolutive Preposing and the special effects of *inggannanna* "all" and *ianasanna* "each'—allows us to reinforce two wider hypotheses about the systems that give rise to the High Absolutive Configuration. First, it must be possible for absolutive arguments to undergo regular steps of movement to dedicated A-positions in the extended TP tightening the parallel with nominative subjects in systems of this type. Second, it must be possible for them to pass through the edge of the clause-internal phase in systems of this type—suggesting that that position may play a special role in the derivation of the High-Absolutive Configuration.

(88) *Clausal Syntax*

a.	SYNTAX:	$[_{\text{TP}}$	DP_{ABS}	v	$[_{\textit{voice}\text{P}}$	DP_{ABS}	DP_{ERG}	$[_{\textit{voice}\text{P}}$	DP_{ABS}	]]].
b.	USUAL PF:	$[_{\text{TP}}$	Δ	v	$[_{\textit{voice}\text{P}}$	Δ	DP_{ERG}	$[_{\textit{voice}\text{P}}$	DP_{ABS}	]]].
c.	SPECIAL PF:	$[_{\text{TP}}$	DP_{ABS}	v	$[_{\textit{voice}\text{P}}$	Δ	DP_{ERG}	$[_{\textit{voice}\text{P}}$	Δ	]]].

The second surrounds the mapping constraints in (89), which mobilize surface prosodic constituency to reflect the input relationships of extended projection and agreement. These constraints are distinct from the standard faithfulness constraints of Match Theory, which demand a more straightforward input-output isomorphism between syntactic and prosodic domains. In wider perspective, however, they fit together with a family of pressures that have been argued to reroute prosodic structure-building to reinstantiate input relationships of many further types, from c-command (Kalivoda, 2018) and selection (Clemens, 2016, 2019) to adjunction (Brodkin, 2025a). To the extent that this investigation is on the right track, then, it provides an additional argument that these pressures exist.

(89) *The Interface Constraints*

a. ATTRACT
   Assign one violation for every functional element that is not parsed into the same phonological phrase as the exponent of its corresponding lexical head.

b. *Group*
   Assign one violation for every absolutive DP and corresponding exponent of ABS agreement that are not parsed into the same maximal phonological phrase.

The third surrounds the architecture of prosodic structure-building along the PF branch. To capture the facts of Absolutive Preposing, I have assumed that prosodic structure is built in the parallel, global, and surface-oriented phonological evaluation of Match Theory. This architectural perspective contrasts with two alternatives that posit some degree of cyclicity in the construction of prosodic domains. The first is the paradigm of CYCLIC SPELL-OUT, which assumes that prosodic domains are built and readjusted as complex syntactic objects are spelled out piece-by-piece (for various proposals, see Dobashi 2004, 2010; Ishihara 2007; Adger 2007; Pak 2008; Newell 2009; Kahnemuyipour 2009). The second is the framework of CONTIGUITY THEORY, which takes prosodic structure-building to proceed in a derivational fashion and allows it to directly influence the syntax (Richards 2016, 2020; see also Richards 2010). In the domain of Chain Reduction, these frameworks diverge around at least the prediction in (90): if there is any type of cyclicity to prosodic structure-building, it should be possible for prosodically conditioned effects to be opaque.

(90) *Predictions about Opacity*

Framework	Cyclicity	Opacity
Match Theory	✗	✗
Cyclic Spell-Out	✓	✓
Contiguity Theory	✓	✓

Outside the arena of Chain Reduction, there is little evidence in the phrasal phonology for opacity of any type (Kiparsky 1985; Booij & Rubach 1987). In the domain of prosodic structure-building, for instance, Cheng & Downing 2016 establish that there is no evidence for cyclicity to be drawn from the distribution of regular phonological phenomena that are sensitive to prosodic phrasing in Chichewa, Kinyambo, Luganda, and Zulu. In Mandar, in the same vein, a reviewer notes that the segmental processes that are sensitive to maximal phonological phrasing seem to afford no special status to the content of the *voice*P phase. To this body of observations we can add one final fact: to derive Absolutive Preposing, it is sufficient to refer to the transparently visible prosodic structure of the surface phonology. In this system of Chain Reduction, in other words, there is no evidence for opacity of any type: the Grouping Constraint is transparently enforced in the surface phonology and the absolutive DP is only spelled out high when that allows it to visibly satisfy that constraint. I leave it to future work to establish whether similar patterns of prosodically conditioned patterns of Chain Reduction are ever able to deviate from this surface-oriented shape.

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